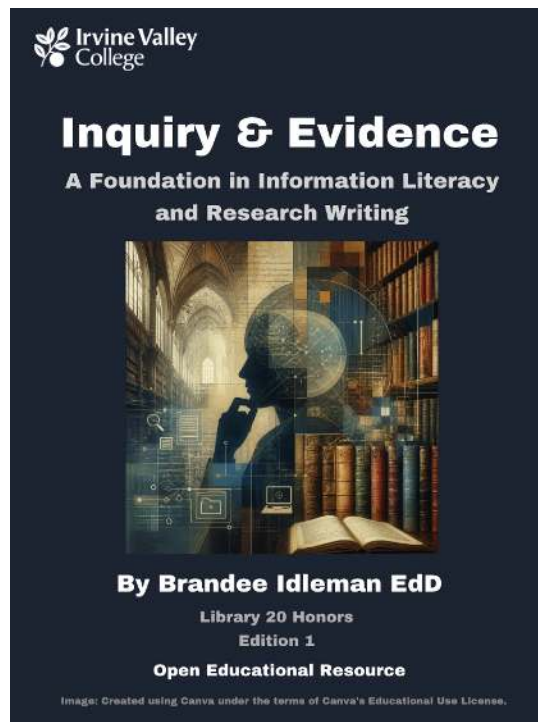


INQUIRY AND EVIDENCE: A FOUNDATION IN INFORMATION LITERACY AND RESEARCH WRITING (IVC)



Brandee Idleman
Irvine Valley College



Advanced Research and Information Competency

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Licensing

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Curriculum Information

Learning Objectives

- Write and create ethically by practicing the use of parenthetical and bibliographic citations.
- Apply organizational structures, like section headings, to written assignments to create cohesion and facilitate audience understanding.
- Demonstrate continued progress in college-level research writing that extracts meaning from existing knowledge and is infused with specific evidence to support findings.
- Write research-infused prose that extracts meaning from scholarly and popular sources.
- Write annotations and critiques of varying information resources and their purpose, including the ability to distinguish objective information and subjective information.
- Write a research question with an articulated research strategy that is flexible and adaptable to accommodate discoveries.
- Apply critical thinking to deconstruct academic articles and reports, identifying common components of research writing and their usefulness.
- Apply critical thinking to iteratively find, select, and access diverse information sources in various contexts and modes, including library databases.

SLOs

1. Students will write a logically organized, evidence-based essay supported by ethical practices that respect copyright, including parenthetical and bibliographic citations.
2. Students will complete a written critique of a scholarly work that contextualizes information to understand its quality, relevance, authority, accuracy, and lens.
3. Students will write a research question that expands on existing scholarly knowledge and is sustainable for undergraduate investigation.
4. Students will write an exploratory essay demonstrating an ability to recognize evidence-based dialog, extract meaning, and draw logical conclusions.

Comparable Transfer Courses

1. Internet and Society
2. Critical Thinking and Composition
3. Argumentative Writing and Critical Thinking

About the Author



About the Author

Dr. Brandee Idleman is a Research & Instruction Librarian and Guided Pathways Coordinator at Irvine Valley College in Southern California. She earned a Doctor of Education from Northeastern University, where her research examined asynchronous information literacy learning in the Canvas learning management system. She also holds a Master of Library and Information Studies from the University of Alabama, where she was inducted into Beta Phi Mu, the international honor society for library and information studies, and a Bachelor of Arts in European History from Dickinson College, where she was a member of Phi Alpha Theta, the national history honor society.

Throughout her career, Dr. Idleman has focused on helping students become confident researchers, critical thinkers, and effective academic writers. She developed Irvine Valley College's asynchronous information literacy program, which has awarded more than 35,000 micro-credentials, and created the college's first honors-level library course, currently approved as a transfer-level critical thinking and research writing course.

As both a librarian and educator, Dr. Idleman teaches research, information literacy, and academic writing while leading campus initiatives that improve student success and equitable access to higher education. This textbook reflects her commitment to making research practical, accessible, and relevant for today's college students.

About This Book

This textbook was written to support **LIB 20H Critical Thinking and Research Writing Honors** at Irvine Valley College. The course develops advanced critical thinking, research, and academic writing through an iterative, evidence-based process. Students learn to formulate meaningful research questions, locate and evaluate scholarly and professional sources, synthesize diverse perspectives, and construct thesis-driven arguments across disciplines. The course culminates in a substantial research project presented as either a conference-ready paper or a multimedia presentation.

While designed for an honors-level research writing course, this book is intended for any student who wants to become a more thoughtful researcher, critical consumer of information, and effective academic writer. Its emphasis on information literacy, evidence-based reasoning, and ethical scholarship provides a foundation for success in college, professional careers, and lifelong learning.

Template

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Learning Objectives

- Objective 1
- Objective 2

jflkjl

Definition: Term

Definition text

lfjljf

The ideal gas law is easy to remember and apply in solving problems, as long as you get the **proper values a**

Caution

The ideal gas law is easy to remember and apply in solving problems, as long as you get the **proper values a**



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Figure 1: Csourced f
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CHAPTER OVERVIEW

1: Foundations of Critical Thinking and Information Literacy

[1.1: What Is Information Literacy and Why Does It Matter?](#)

[1.2: Critical Thinking, Logic, and Rhetoric](#)

[1.3: Context, Positionality, and the Researcher's Lens](#)

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1.1: What Is Information Literacy and Why Does It Matter?

Unit 1 Foundations of Critical Thinking & Information Literacy|Chapter 1.1

Learning Objectives

- Define information literacy and explain why it matters in academic and everyday contexts.
- Distinguish among academic, media, news, digital, AI, civic, personal, health, and financial literacies.
- Identify the five components of information literacy: identify, find, evaluate, apply, and acknowledge.

Introduction

Why Information Literacy Matters

You make information decisions every day, from choosing which TikTok to watch to deciding which side of a political argument deserves your attention. **Information literacy is not just a library skill; it is a survival skill.** The world runs on information, and the people who learn to navigate it well end up with more options, more leverage, and greater peace of mind than those who don't.

Most people assume that "being informed" means having more information. It doesn't. There has never been more information available, yet people are not collectively making better decisions. The bottleneck is no longer quantity. It is the quality of your attention, the rigor of your evaluation, and your willingness to slow down and ask, *How do I actually know this?* This book is about that slowdown.

Defining Information Literacy

Throughout this book, you will become increasingly information literate. What does that mean? Broadly, being information literate means you can locate and select reliable sources of information in all its forms, including text (books, articles) and media such as audio or visual resources like photographs and videos. One information resource — a website, for example — may require you to evaluate multiple forms of information at once. If you visited the Wall Street Journal's website, you'd find text articles, videos of news stories, and photographs all on the same page.

Being information literate is only becoming more complex. Every day, we process massive amounts of information in various forms, often discriminating between types of information without consciously evaluating them. Within academics, information literacy is essential. The Association of College and Research Libraries (ACRL, 2015) defines it as a set of integrated abilities encompassing the reflective discovery of information, the understanding of how information is produced and valued, and the use of information in creating new knowledge and participating ethically in communities of learning.

Multiple Types of Information Literacy

"Information literacy" is closer to a family of related skills than a single one. Knowing the family helps you recognize which kind of literacy a situation needs.

There is no definitive list of the types of information literacy. Most scholars agree on a core set: media, digital, data, visual, and information ethics. Beyond that core, the family stretches to fit the situations of your life and may include health, financial, news, civic, personal, and AI literacies, among others. The table below describes the core five; the situational literacies appear throughout this book.

Table 1.1 The core information literacies and the knowledge and skills each contributes to evaluating, using, and communicating information effectively.

The Core Information Literacies	
Media Literacy	Critically analyze news, entertainment, and social media by recognizing purpose, bias, framing, and the difference between fact, opinion, and advertising.
Digital Literacy	Effectively navigate digital environments by using technology, searching for information, managing files, protecting personal information, recognizing online threats, and understanding how algorithm online experiences.
Data Literacy	Interpret, evaluate, and communicate data by understanding statistics, sampling, visualizations, and can—and cannot—support as evidence.
Visual Literacy	Interpret, evaluate, and create meaning from images, charts, videos, and other visual media while recognizing how visuals persuade, inform, and can be manipulated.
Information Ethics	Use information responsibly by citing sources, respecting intellectual property and privacy, protecting personal data, and applying information ethically and honestly.

Emerging Information Literacies

As the information landscape evolves, several core and situational literacies have taken on greater significance and deserve closer examination.

Media literacy is the broad skill of reading any mediated message with your eyes open. Every message you encounter was constructed: written, framed, edited, scored, and cropped by people (or algorithms) with purposes of their own. Media literacy asks who created a message, what techniques it uses to hold your attention, what it leaves out, and who benefits when you believe it. In an attention economy, where feeds are engineered to keep you scrolling, those questions have never mattered more.

News literacy is media literacy applied to journalism: the ability to evaluate news sources well enough to decide what to trust. It includes telling reporting from opinion, recognizing misinformation (false content shared by mistake) and disinformation (false content built to deceive), and verifying a story before you share it. Chapter 4.2 is devoted to these skills, including the CRAAP test, lateral reading, and triangulation.

Artificial intelligence literacy is the newest member of the family. It means understanding, at a working level, what generative AI tools can and cannot do: they produce fluent, confident text and images, but they also invent facts and citations, reflect the biases of their training data, and verify nothing. AI literacy also means recognizing that content you encounter may be synthetic. Deepfakes, meaning AI-generated audio, images, or video that convincingly impersonate real people, can put words in a politician's mouth or fabricate an event that never happened. The defenses are the habits this book teaches: check the source, read laterally, and treat any striking claim that appears in only one place as unverified (Chapter 4.2). Using AI tools ethically in your own work is covered in Chapter 6.1.

These literacies overlap. Deciding whether to take out a student loan is a financial decision, but it also requires media, digital, civic, and personal literacy. Strengthening one strengthens the others.

Civic literacy in particular has deep roots in the library profession. Writing in Library Journal fifty years ago, librarian and future U.S. Congressman Major Owens (1976) argued that democratic societies face a crucial information deficit: citizens cannot govern themselves without systems that make information about government visible, accessible, and usable. He predicted that political involvement would become a bread-and-butter necessity of everyday life and that the demand for open government and access to official documents would keep growing. His forecast describes the world you now live in, where civic participation depends on the ability to find, evaluate, and use information about how you are governed.

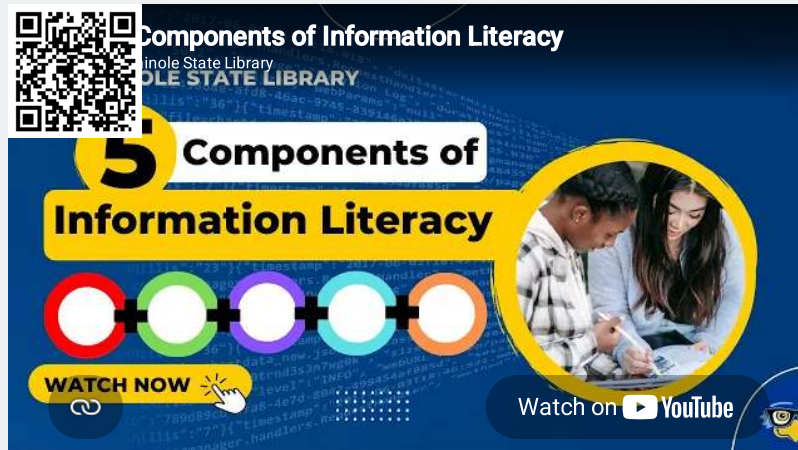
The Five Components of Information Literacy

Information literacy is a sequence of moves you repeat, in different orders, for the rest of your life. Throughout this book we will use the widely taught five-component framework summarized in the short video below: identify, find, evaluate, apply, and acknowledge (Seminole State Library, n.d.).

1. **Find:** Locate sources where credible answers might live (databases, catalogs, government data).
2. **Evaluate:** Judge each source for authority, accuracy, currency, relevance, perspective, and purpose.
3. **Apply:** Read, integrate, and use the source to quote, paraphrase, summarize, or build an argument.
4. **Acknowledge:** Cite sources so your reader can follow your trail and credit the original creators.

5 Components of Information Literacy

Watch this 5 minute video. Captions are available in the video player.



Why it's a cycle, not a checklist

These five components rarely happen in a clean line. You will find a source that changes your question. Evaluating sources will send you back to find more. When your work loops back on itself, you are not failing, you are doing real research.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 1.1A — 24-Hour Information Audit

For 24 hours, log every time you intentionally consult information. Note the source, the question, and whether it was the best source you could have used. Then write a 200-word reflection on which literacies you leaned on most and where you cut corners.

Activity 1.1B — Five Components, One Decision

Think of a real decision you made recently that required information: choosing a class, making a significant purchase, changing a habit. Walk it through the five components. What did you need to know (identify)? Where did you look (find)? How did you judge what you found (evaluate)? How did you use it (apply)? Could you retrace your sources if someone asked (acknowledge)? Write a 150-word reflection on which component you did well and which one you skipped.

? Think About It: Discussion Questions

1. Of the many literacies, which one is hardest for you in academic work? Why?
2. Owens (1976) argued that democracy fails without informed citizens. Think of a local decision that affects you directly. What information would you need to participate in it meaningfully, and where would you find that information?

Conclusion

What's Ahead

Every chapter that follows returns to the five components above, putting one move under the microscope at a time. Chapter 1.2 pairs information literacy with its companion skill, critical thinking, and gives you tools for taking arguments apart. Unit 2 builds your topic and research question. Unit 3 sharpens your search skills. Unit 4 teaches you to read and evaluate scholarly sources. Unit 5 walks you through synthesis and the literature review. Unit 6 closes the loop with ethical citation and research presentation. Keep this chapter handy; its vocabulary will reappear in later chapters.

Going Further: The Frames Behind the Framework

This chapter used the five-component model because it is practical and easy to remember, but the [ACRL Framework](#) (2015) that anchors college information literacy is built on six deeper frames, including Authority Is Constructed and Contextual and Scholarship as Conversation. For an honors extension, [read the Framework itself](#) (it is short and freely available online). You will meet all six frames in the chapter alignment tables throughout this book.

Chapter Summary

Information literacy is the ability to find, select, and use high-quality information that is appropriate to your purpose (ACRL, 2015). It is best understood as a family of related literacies (academic, media, news, digital, AI, civic, personal, health, and financial), and it operates through five components: identify, find, evaluate, apply, and acknowledge (Seminole State Library, n.d.). The chapters that follow return to these moves in deeper and deeper ways, beginning with Chapter 1.2, which introduces information literacy's companion skill: critical thinking.

Key Concepts and Vocabulary

- **Information Literacy.** The set of integrated abilities for finding, evaluating, using, and acknowledging information appropriate to your purpose (ACRL, 2015).
- **Academic literacy.** Finding, reading, and using scholarly sources such as peer-reviewed articles and books.
- **Media literacy.** Reading news and entertainment with attention to framing, omission, and the line between news and opinion.
- **News literacy.** Evaluating news sources well enough to decide what to trust, including telling reporting from opinion and spotting false content.
- **AI literacy.** Understanding what generative AI tools can and cannot do, recognizing synthetic content such as deepfakes, and using AI ethically.
- **Civic literacy.** Making sense of political and civic information well enough to participate in self-governance.
- **Information deficit.** The gap between the information a democratic society needs and the information that actually reaches its citizens. (Owens, 1976.)

Attributions and References

Association of College and Research Libraries. (2015, February 9). *Framework for information literacy for higher education*. American Library Association. <https://www.ala.org/acrl/standards/ilframework>

Owens, M. R. (1976, January 1). The state government & libraries. *Library Journal*, 101(1), 19–28.

Seminole State Library. (n.d.). *5 components of information literacy* [Video]. YouTube. <https://www.youtube.com/watch?v=1ronp6lue9w>

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1.2: Critical Thinking, Logic, and Rhetoric

Unit 1 Foundations of Critical Thinking & Information Literacy | Chapter 1.2

Learning Objectives

- Define critical thinking and explain how it drives information literacy.
- Use Bloom's revised taxonomy to describe the cognitive moves involved in research and writing.
- Distinguish among logic, reasoning, and rhetoric, and explain how they overlap.
- Identify Aristotle's three rhetorical appeals — ethos, pathos, and logos — in a real-world argument.
- Use the eight-element logic model to take an argument apart and to build one of your own.
- Recognize when rhetorical strategy is being used ethically and when it is being used to manipulate.

Introduction

Critical Thinking: The Engine of Information Literacy

Information literacy is more than finding information. It requires thinking critically about the information you encounter, questioning its credibility, evaluating the quality of evidence, recognizing assumptions and bias, and drawing reasoned conclusions. In other words, **critical thinking is the engine that powers information literacy**. Without it, information literacy becomes little more than locating and collecting sources.

Critical thinking is the ability and habit of objectively analyzing, evaluating, and synthesizing information, ideas, and arguments to make informed decisions. Rather than accepting information at face value, critical thinkers ask questions, consider multiple perspectives, weigh evidence, and reflect on the implications of their conclusions. These habits are essential for academic research and equally valuable in everyday life, whether evaluating a news story, interpreting data, making personal decisions, or participating in civic discussions.

Critical thinking is also transferable. It is not limited to research writing or any single discipline. Whether you are studying biology, history, business, or engineering, the same habits of careful analysis, evidence-based reasoning, and intellectual humility apply. Strong critical thinkers recognize the limits of their own knowledge, remain open to new evidence, and are willing to revise their conclusions when the evidence warrants it. These qualities make them not only better researchers but also more informed citizens and lifelong learners.

Bloom's Taxonomy: A Map for Thinking

If critical thinking is a cluster of habits, Bloom's Taxonomy is a map of the mental moves those habits ask you to make. Originally introduced by Benjamin Bloom in the 1950s and revised by Anderson and Krathwohl in 2001, it describes six levels of cognitive work, arranged from foundational to advanced (Armstrong, 2010).

A taxonomy is a framework for organizing knowledge or concepts into meaningful categories.. Bloom's taxonomy gives you vocabulary for the kind of thinking different assignments require, and it shows that thinking has a sequence — you cannot effectively evaluate or create without first remembering, understanding, and applying.

Bloom's Taxonomy

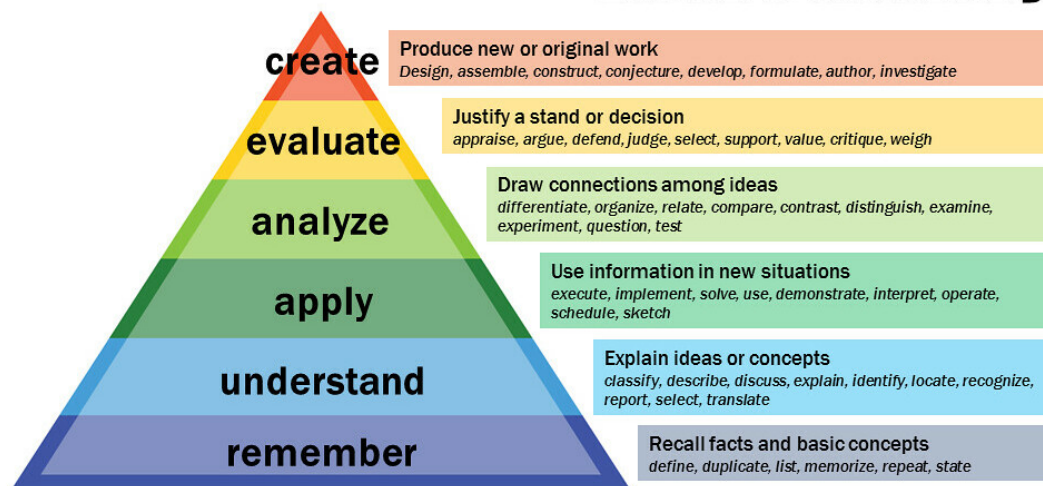


Figure 1.2.1: Diagram of Bloom's Taxonomy showing six cognitive levels: Remember, Understand, Apply, Analyze, Evaluate, and Create.. (Copyright; <http://cft.vanderbilt.edu/guides-sub-pages/blooms-taxonomy/>)

Most research projects move through Bloom's levels in roughly this order: You begin by remembering and understanding (What is this topic about?), shift to applying and analyzing (How do these sources relate?), and finish at the upper levels by evaluating sources, drawing conclusions, and creating something original, such as an essay, literature review, or presentation. A literature review is more demanding than a summary because it operates at Bloom's higher levels (analyze, evaluate, create) rather than the lower levels (remember, understand).

✓ Example 1.2.1

Bloom's Taxonomy in Everyday Life

Consider the steps involved in learning to drive a car.

- **Remember:** Learn the meaning of traffic signs and road rules.
- **Understand:** Recognize what traffic signs and road markings mean in different driving situations.
- **Apply:** Drive on neighborhood streets using those rules.
- **Analyze:** Recognize why another driver made a mistake and how traffic conditions affect your decisions.
- **Evaluate:** Decide which route is safest or whether another driver's actions were appropriate.
- **Create:** Plan a road trip by choosing the best route, anticipating challenges, and adapting your driving to changing conditions.

Research works the same way. You begin by learning and understanding information, then apply and analyze it, evaluate the quality of evidence, and finally create an original argument or solution.

Critical Thinking in Action

The previous sections defined critical thinking and introduced Bloom's Taxonomy as a framework for understanding the different levels of thinking involved in research. The next step is learning how to apply those thinking skills to the information you encounter every day.

In this section, you'll learn how to analyze messages designed to persuade you, whether they come from a news editorial, a campaign ad, a research article, a TikTok post, or an email from your bank. Learning to recognize how arguments are constructed will help you become a more thoughtful reader and a stronger writer.

✓ Example 1.2.2

Practicing Critical Thinking

Imagine you read the headline, "*Students Who Drink Coffee Earn Higher Grades.*" Before accepting the claim, ask a few critical thinking questions:

- What evidence supports this claim?
- Does the article confuse correlation with causation?
- Who conducted the study, and who funded it?
- Are there alternative explanations for the results?

These are the kinds of questions critical thinkers ask before accepting an argument or using it as evidence in their own research.

Two concepts make the process of applying critical thinking easier. The first is **rhetoric**, the art of using language to persuade. The second is the **logic model**, the underlying structure that holds an argument together. Rhetoric helps you understand *how* a message persuades; the logic model helps you evaluate *whether the argument is logically sound*. Strong researchers learn to examine both before deciding what to believe or cite.

Rhetoric: How Language Shapes What You Believe

Rhetoric is the art and skill of using language effectively and persuasively to communicate, convince, influence, or motivate an audience. It encompasses a range of techniques and strategies designed to shape how people perceive and respond to information, ideas, and arguments. Rhetoric is not simply about conveying information—it is about presenting information in ways that encourage a particular response or action.

That last word—**response**—matters. Rhetoric itself is neutral. The same techniques that help a public health campaign save lives can also be used by a scam artist to steal your money. Learning rhetoric is one of the best ways to recognize when persuasive techniques are being used ethically—and when they are being used to manipulate or deceive.

The three classical appeals: ethos, pathos, logos

Aristotle, the ancient Greek philosopher, identified three primary modes of persuasion that still describe how humans try to convince each other today (Aristotle, ca. 350 BCE/1991).

The Three Classical Rhetorical Appeals: Ethos, Pathos, and Logos

Ethos, Pathos, Logos	
Ethos	Appeals to the speaker's credibility, authority, and character. "Trust me, I'm a doctor."
Pathos	Appeals to the audience's emotions: fear, hope, anger, pride, sympathy. Charity ads lean heavily on pathos.
Logos	Appeals to logic and reason: evidence, data, structured argument, cause-and-effect. Most peer-reviewed scholarship leads with logos.

Real arguments rarely use just one appeal. A great speech blends all three: the speaker's credibility (ethos), our emotional connection to the topic (pathos), and the evidence they walk us through (logos). When you read or watch something persuasive, ask yourself: which appeals are doing the heavy lifting? If the argument leans almost entirely on emotion, that is a signal to slow down.

The other moves rhetoric makes

Beyond the three appeals, rhetoricians use audience awareness (knowing what their listeners already believe and care about), rhetorical devices (metaphors, similes, hyperbole, alliteration, parallelism), figures of speech (analogy, irony, allegory, personification), and delivery (tone, rhythm, pacing). All of these affect how persuasive a message feels — separate from whether the underlying claim is actually true.

The Logic Model: Inside an Argument

If rhetoric is the dress, the logic model is the skeleton. A logic model is a structured framework that outlines the relationships among the components of an argument. The model below builds on philosopher Stephen Toulmin's classic argumentation framework (Toulmin, 1958/2003). It serves two purposes: it helps you break down other people's arguments, and it helps you construct your own arguments so they are logically sound.

The Eight-Element Logic Model for Analyzing and Constructing Effective Arguments

The Eight-Element Logic Model	
1. Claim	The main proposition or assertion the communicator is trying to prove. The center of gravity of the argument.
2. Evidence	Factual information, data, examples, expert opinions, or anecdotes that support the claim.
3. Reasoning	The logical connections that link the evidence to the claim — the "because" between them.

The Eight-Element Logic Model

4. Warrants	Underlying assumptions or principles that justify treating the evidence as relevant. Often unstated.
5. Backing	Additional support that reinforces the warrants and shows they are reasonable.
6. Counterarguments	Potential objections or opposing viewpoints the communicator anticipates from the audience.
7. Rebuttals	Responses that address those counterarguments and explain why they don't undermine the main claim.
8. Conclusion	The final statement that summarizes the argument, restates the claim, and reinforces the reasoning.

Notice how much hidden work warrants do. *Smoking causes cancer* (claim) is supported by decades of epidemiological data (evidence), but the warrant—that statistical correlation across millions of people, controlled for other variables, justifies a causal inference—is rarely stated explicitly. Strong critical thinkers learn to identify warrants and ask whether they are justified.

Also notice items 6 and 7. A strong argument does not pretend opposing views do not exist. Instead, it acknowledges them and responds to them. When an argument ignores counterarguments altogether, that is a signal in itself: either the writer does not understand the broader conversation, or they are hoping you do not.

Quick Test

Where is the argument weakest?

When you read something persuasive, find the eight elements. The element that's missing, or the one that's there but thin, is the weakest point in the argument. That's where you should focus your skepticism, and where your own writing should focus its strongest support.

Critical Thinking Activities

? Try It: Reflection Activity

Activity 1.2A — Read an Op-Ed Like a Critical Thinker

Choose a recent opinion piece from a credible newspaper. First, identify each of Aristotle's three appeals: where does the writer use ethos? pathos? logos? Then, map the eight-element logic model: what is the claim? the evidence? the reasoning? the warrants? Are counterarguments acknowledged and addressed? Where does the argument feel strongest, and where does it feel thin?

Activity 1.2B — Bloom's Self-Diagnosis

Pull up your most recent assignment in any subject. Underline every verb in the prompt and match each to one of Bloom's six levels. Write 150 words on what level the prompt is really asking for and what would change about your draft if you wrote it at one level higher.

? Think About It: Discussion Questions

1. Think of an argument you've recently been persuaded by. Looking back, which of Aristotle's three appeals — ethos, pathos, logos — did most of the work? Are you comfortable with that?
2. "Rhetoric is neutral." Do you agree? When does a powerful rhetorical technique cross the line into manipulation?

Conclusion

Why This Matters for Research

Every research paper is an argument. Every scholarly source you'll read this semester is somebody's attempt to persuade you that a claim is true based on evidence. The chapters ahead will teach you to find sources, evaluate them, synthesize them, and cite them — but underneath all of that is the work this chapter introduced: noticing rhetoric, mapping logic, and asking what a piece of writing is really doing to your brain.

Bloom's Taxonomy puts the moves you've learned here at the upper levels of cognitive work: analyzing arguments (Level 4) and evaluating them (Level 5). That is the level the rest of this book will ask you to operate at.

Going Further: Qualifiers and the Strength of Claims

Toulmin's original model includes an element this chapter's adaptation folds into reasoning: the qualifier, the small words (most, often, in many cases, under these conditions) that mark how strong a claim is meant to be. Strong scholarly writing is full of qualifiers; weak persuasion tends to drop them. For an honors extension, take an op-ed and a peer-reviewed article on the same topic and compare their qualifiers. What does the difference tell you about how each writer treats certainty?

Chapter Summary

Critical thinking is the cognitive skill and mindset of analyzing, evaluating, and synthesizing information to form reasoned judgments rather than accepting claims at face value. Bloom's revised taxonomy (Armstrong, 2010) maps that work across six levels, from remembering and understanding up to evaluating and creating. Critical thinking becomes concrete when you have tools for taking arguments apart. Rhetoric, the art of persuasion, relies on three classical appeals (ethos, pathos, logos) along with audience awareness, rhetorical devices, and delivery. The logic model gives you the underlying skeleton of an argument: claim, evidence, reasoning, warrants, backing, counterarguments, rebuttals, and conclusion. Both frameworks let you read other people's writing more carefully and write your own arguments with more rigor.

Key Concepts and Vocabulary

Critical thinking. The skill and mindset of analyzing, evaluating, and synthesizing information to form reasoned judgments rather than accepting claims at face value.

Bloom's revised taxonomy. Six levels of cognitive work, from remember and understand up to evaluate and create (Anderson & Krathwohl, 2001).

Rhetoric. The art and skill of using language persuasively to shape how an audience perceives and responds.

Ethos, pathos, logos. Aristotle's three appeals: to the speaker's credibility, to the audience's emotions, and to logic and evidence.

Logic model. An eight-element framework for the structure of an argument, adapted from Toulmin: claim, evidence, reasoning, warrants, backing, counterarguments, rebuttals, and conclusion.

Warrant. The underlying, often unstated assumption that makes evidence count as support for a claim.

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1.3: Context, Positionality, and the Researcher's Lens

Unit 1 Foundations of Critical Thinking & Information Literacy | Chapter 1.3

Learning Objectives

- Define positionality and explain why it matters for research.
- Distinguish perspective from bias.
- Identify the elements of your own researcher's lens — identity, lived experience, disciplinary training, and prior beliefs.
- Draft a positionality statement for your own emerging research interest.

Introduction

What Is Positionality?

To be an objective researcher and writer, you must be self-aware. That sentence sounds like a paradox. *How can you be objective and self-aware at the same time?* The answer is the concept of positionality.

Positionality refers to the social, cultural, and personal location of a researcher. This includes the identities, experiences, and beliefs that shape how they see a problem, what questions they ask, and how they interpret evidence (Bourke, 2014; Holmes, 2020). Every researcher has a position. The question is whether they are aware of it. Researchers who acknowledge their positionality produce work that is more rigorous, not less, because the reader can see what is influencing the analysis.

There is no "view from nowhere" in research. The myth of the perfectly neutral observer ignores the simple fact that every observer is a person — a person with a body, a history, an education, a generation, a worldview. Pretending otherwise hides those influences instead of accounting for them.

Perspective vs. Bias

Perspective and bias are easy to confuse, and the distinction matters. **Perspective is a point of view.** All research, like all news reporting, comes with some perspective because it is created by humans. It is impossible to completely remove your point of view; what serious researchers do instead is strive for objectivity while disclosing what is shaping their lens.

Bias is something different. **Bias is a prejudice or inclination** to read evidence that reflects personal or institutional beliefs and lacks objectivity. Biased work ignores facts that don't fit the desired conclusion or exaggerates the facts that do. Where perspective is unavoidable, bias is a problem to be checked.

Positionality work is how you keep your perspective from quietly hardening into bias. By naming the lens you bring, you make it visible — to your reader and to yourself — and that visibility is the discipline that allows you to reach for fair conclusions.

Mapping Your Lens

Your positionality is made of several overlapping layers. You don't need to dissect every one of them for every project, but knowing the layers exist helps you notice when one of them is steering your work.

The Four Layers of a Researcher's Lens That Shape Perspective and Interpretation

Four Layers of a Researcher's Lens	
Identity	Age, gender, race, ethnicity, culture, language, sexuality, ability, religion, nationality, class. Identity may include categories society has used to sort you and that you may or may not embrace yourself.
Lived experience	Where and how you grew up; your family, schooling, work, relationships; the places you have or haven't lived; the systems you have or haven't navigated.
Disciplinary training	The fields whose habits of thought you have absorbed — psychology, business, nursing, art, engineering — and the kinds of evidence those fields privilege.
Prior beliefs	The assumptions you carried into the topic before you started reading. What did you already think you knew? What do you suspect was wrong?

These layers shape the questions you find interesting, the sources you trust, the experts you cite, and the conclusions that feel persuasive to you. They are not flaws to be eliminated. They are inputs to be acknowledged.

✓ Example 1.3.1

Researcher Perspective

Two researchers study the same question: "How does access to public transit affect college persistence?" One grew up in a city, never owned a car, and rode the bus to high school. The other grew up in a rural area where public transit didn't exist. Same question, two starting lenses. Both can do excellent work, but neither will do their best work pretending they walked into the question from nowhere.

Why Scholars Write Positionality Statements

Across the social sciences, education, public health, and humanities, scholars increasingly include a short positionality statement in their published work. The statement names who the researcher is in relation to the topic, what experiences they bring, and how those experiences may have shaped their interpretation. The convention started

in qualitative research methodologies where the researcher is the primary instrument of analysis, and it has spread because reviewers and readers have come to value the transparency it provides (Holmes, 2020).

A positionality statement is not a confession or a disclaimer. It is information your reader needs in order to evaluate your work. It signals that you are aware of your own influence on the analysis and that you have taken steps to keep that influence honest.

Drafting Your Positionality Statement

A solid positionality statement for an early research project answers four questions:

1. Who are you in relation to this topic? What identities and experiences place you near it or distant from it?
2. What assumptions did you bring with you when you started? What did you think you already knew?
3. What has changed as you've read? Where has the evidence pushed back on what you thought?
4. What are you watching out for? Where might your lens still be steering your interpretation?

For this class, **you will write a positionality statement in the first person**. However, in journal articles, it is typical to use first person and integrate the statement within the study's limitations.

✓ Example 1.3.2

Excerpt of Third-Person Positionality Statement

The researcher has little influence in decision-making at the governance level but is engaged in high-profile committee work that offered a means to connect with stakeholders across campus. The researcher was in a lesser position among peer faculty due to a lack of tenure status. It was necessary to manage faculty demands professionally and diplomatically that may not best represent students' needs. As an insider with students, the researcher could be viewed as an authority figure needing to be pleased. Even in circumstances where the researcher was not explicitly connected to student grades, an aura of authority comes with being a teacher. Throughout the study, the researcher was sensitive to the classroom hierarchy.

✓ Example 1.3.3

Excerpt of First-Person Positionality Statement

I was not in a position to make governance-level decisions, but my involvement in several high-profile campus committees allowed me to build relationships with stakeholders across the college. At the same time, I was a non-tenured faculty member, which meant I often had to navigate competing priorities and respond diplomatically to the expectations of colleagues, even when those expectations did not always align with what I believed was best for students.

As both a researcher and a faculty member, I recognized that students might view me as an authority figure, even when I had no role in grading them. Because of this, I remained mindful of the power dynamics that naturally exist between faculty and students and worked to create an environment where participants felt comfortable sharing their honest perspectives without feeling pressured to please me.

Be Specific

When you write your statement, be specific. Avoid vague gestures like "I have always been interested in this topic" because those don't help your reader. The most useful statements name something concrete: a job you held, a community you belonged to, a honest examination of your social media consumption, how your family or friends have influenced your beliefs and assumptions, a moment when your view changed.

📌 Quick Test

After you finish drafting your positionality statement, share it with someone who is unfamiliar with your topic. If they can describe your perspective in one sentence, your statement is working. If they say, *"I'm not sure what your perspective is on this,"* go back and add more specific details.

Conclusion

Going Further: Reflexivity in Published Research

Scholars call the ongoing practice of examining your own influence on your research reflexivity, and qualitative methodologists have built formal techniques for it, including reflexive journals and memo writing. For an honors extension, find a qualitative study in a field that interests you and trace how (or whether) the authors account for their own positionality. Holmes (2020) is a good starting point for the vocabulary researchers use.

Chapter Summary

Positionality is the social and personal location every researcher brings to their work. Acknowledging it makes research more credible, not less, because it makes the influences on your analysis visible. Perspective is unavoidable; bias is what perspective becomes when it goes unexamined. By mapping the layers of your own lens — identity, lived experience, disciplinary training, prior beliefs — and writing a clear positionality statement, you bring the discipline of self-awareness into the foundation of every research project you start.

🔑 Key Concepts and Vocabulary

- **Positionality.** The social, cultural, and personal location a researcher brings to their work, shaping the questions they ask and how they interpret evidence.
- **Perspective.** A point of view; unavoidable in all information created by humans.
- **Bias.** A prejudice or inclination that lacks objectivity, ignoring facts that do not fit a desired conclusion.
- **Researcher's lens.** The combination of identity, lived experience, disciplinary training, and prior beliefs through which a researcher sees a topic.
- **Positionality statement.** A short first-person statement naming who the researcher is in relation to the topic and how that may shape the analysis.
- **Reflexivity.** The ongoing practice of examining your own influence on your research as it unfolds.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 1.3A — Identity Wheel

On a blank page, draw a circle. Around it, write eight identities or experiences that shape how you see the world (e.g., first-generation student, parent, military veteran, transit rider, psychology major, etc.). For each, write one sentence on how that identity might shape how you'd approach a research topic in your area of interest. There is no wrong answer; this is a thinking exercise.

Activity 1.3B — Find a Published Positionality Statement

Search a database (Academic Search Complete, ProQuest, or Google Scholar) for a peer-reviewed article that includes a positionality statement. Copy the statement and write a 100-word response: what does the author disclose, and how does it change how you read the rest of their work? If you cannot find a positionality statement, ask your professor for their statement.

? Think About It: Discussion Questions

1. Have you ever read something, such as a news story, a study, or an opinion piece, and thought, "This person clearly has an axe to grind"? What signaled that to you? Was it perspective, or had it crossed into bias?
2. Which of the four layers of the researcher's lens (identity, lived experience, disciplinary training, and prior beliefs) do you think shapes your perspective most strongly? Which layer is hardest for you to recognize in yourself?

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CHAPTER OVERVIEW

2: Exploring the Information Landscape

- [2.1: Understanding the Information Universe](#)
- [2.2: Exploratory Research](#)
- [2.3: Refining a Research Topic](#)
- [2.4: Crafting a Sustainable Research Question](#)

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2.1: Understanding the Information Universe

Unit 2 Exploring the Information Landscape| Chapter 2.1

Learning Objectives

- Describe the information cycle and explain why the timing of an information source affects how it should be used.
- Distinguish among objective, subjective, propaganda, satire, parody, and opinion content.
- Apply three useful source pairs: objective vs. subjective, scholarly vs. popular, and primary vs. secondary.
- Match a research need to the appropriate source type — database, catalog, government data, news, or media.

Introduction

Every day, people encounter information through news stories, social media posts, videos, podcasts, websites, books, and scholarly research. Although these sources look very different, they all play a role in how knowledge is created and shared.

This chapter introduces three foundational ideas that every researcher should understand:

- How information develops over time.
- The different forms information can take.
- How to recognize the type of source you are using.

Understanding these concepts will help you choose appropriate evidence, avoid common research mistakes, and become a more informed consumer of information.

The Information Cycle

After a newsworthy event occurs, information becomes available in a predictable pattern called the **information cycle**. Each stage of the cycle serves a different purpose and answers different kinds of questions. Knowing where your topic falls in the information cycle helps you select the most appropriate sources for your research (University of Illinois Urbana-Champaign University Library, n.d.).

Video Title

Watch this 3 minute video. Captions are available in the video player.



The Information Cycle

Libraries



Watch on YouTube

How Information Becomes Available

1. **The day an event occurs**, information spreads rapidly through television, live news coverage, social media, blogs, and websites. Speed is the priority. Reports are often incomplete, details may change, and errors are common.
2. **During the first week**, newspapers begin producing more complete stories. Journalists interview witnesses, verify facts, and provide additional context. Although corrections may still be needed, reporting becomes more reliable.
3. **Several weeks later**, magazines and long-form news organizations publish deeper analyses. Rather than simply describing what happened, they explain why it happened and explore its broader significance.
4. **Months later**, scholars begin publishing peer-reviewed research. Academic studies take much longer because experts review the work before publication. These sources emphasize carefully documented evidence instead of immediate reporting.
5. **After a year or more**, books, government reports, encyclopedias, and other reference works synthesize information from many sources. These publications provide the historical record and place events into a broader context.

The information cycle matters because not every type of source exists immediately after an event. If you are researching something that happened last week, you probably will not find scholarly articles yet. If you are researching an event from twenty years ago, you will likely have access to every stage of the information cycle.

Choosing sources that match your topic's place in the information cycle will save time and help you find stronger evidence.

Types of Information

Not all information serves the same purpose. Before evaluating a source, first identify what type of information you are reading.

Six common types of information and how they differ in purpose and use.

Six Types of Information		
Type	Description	Examples / Notes
Objective Information	Presents facts that can be verified through evidence; attempts to minimize personal opinion and bias.	Most scholarly research strives to be objective.
Subjective Information	Includes personal interpretation, analysis, emotions, or judgment; not necessarily inaccurate, but reflects a particular perspective.	Personal analysis and reflections.

Six Types of Information

Opinion	Argues for a particular viewpoint; may include facts and evidence, but its primary purpose is persuasion.	Editorials, columns, many blog posts.
Propaganda	Selectively presents or distorts information to influence beliefs or behavior; attempts to shape public opinion rather than encourage critical thinking.	Often political, ideological, or religious messaging.
Satire	Uses humor, irony, or exaggeration to criticize people, institutions, or society.	<i>The Onion</i> intentionally creates fictional stories that imitate r
Parody	Imitates a specific person, event, or work for humorous or critical effect.	Editorial cartoons, comedic impersonations, spoof videos.

Selecting Information for Research

Most academic research relies primarily on objective evidence, much of which comes from peer-reviewed journal articles. Depending on the discipline and research question, those articles may be primary or secondary sources. Subjective and opinion sources can also be valuable when examining public attitudes, persuasive arguments, or cultural perspectives. Satire, parody, and propaganda are generally not used as factual evidence. Instead, they are analyzed as examples of communication, persuasion, or popular culture.

Three Important Distinctions

Several pairs of concepts are commonly confused by new researchers. Understanding the differences will help you choose sources more effectively.

Objective vs. Subjective

Objective sources emphasize evidence that can be independently verified. Subjective sources emphasize personal interpretation or judgment.

Most academic research relies primarily on objective evidence, much of which comes from peer-reviewed journal articles. However, the important question is whether the source matches your research purpose. A historian studying public opinion may intentionally use subjective sources, while a scientist reporting experimental results should rely primarily on objective evidence.

Scholarly vs. Popular

Students often confuse scholarly and popular sources because both can discuss the same topic.

Popular sources, including newspapers, magazines, podcasts, and news websites, are written for general audiences. They use accessible language, are published quickly, and help readers stay informed about current events or gain introductory knowledge.

Scholarly sources are written by experts for other experts. Most undergo peer review before publication and follow a predictable structure that often includes an abstract, introduction, methods, results, discussion, and references. They use specialized vocabulary and focus on contributing new knowledge to a field.

Most academic research relies primarily on objective evidence, much of which comes from scholarly peer-reviewed journal articles. However, popular sources are often the best choice for understanding current events or gaining background information. Scholarly sources are generally preferred when conducting academic research because they provide deeper analysis and carefully documented evidence.

Primary, Secondary, and Tertiary Sources

Researchers also classify sources according to their relationship to an event or idea.

- **Primary sources** provide firsthand evidence. Examples include diaries, interviews, photographs, speeches, original research data, legal documents, creative works, and eyewitness accounts.
- **Secondary sources** interpret, analyze, or explain primary sources. Most scholarly journal articles, biographies, documentaries, and history books are secondary sources because they examine existing evidence rather than presenting firsthand observations.
- **Tertiary sources** summarize information from many primary and secondary sources. Encyclopedias, textbooks, handbooks, and topic overviews help readers build background knowledge before conducting deeper research.

Common Misconception

Students often assume that secondary sources are less reliable than primary sources. In reality, many of the most credible scholarly articles are secondary sources because they analyze and interpret primary evidence. The terms **primary**, **secondary**, and **tertiary** describe a source's relationship to the evidence. They do not indicate whether a source is trustworthy or high quality.

Where to Search for Information

Knowing **where to search** is just as important as knowing **what to search for**. Different search tools are designed to help you find different kinds of information. Choosing the right place to begin can save time and lead to better research results.

Library Databases

Library databases provide access to high-quality resources that are selected and licensed by the library. Depending on the database, you may find scholarly journal articles, newspapers, magazines, ebooks, streaming media, reports, primary sources, and other specialized content.

Examples include **Academic Search Complete**, **ProQuest Research Library**, **JSTOR**, **Newspaper Source Plus**, and **Nexis Uni**.

Library Catalog

The library catalog helps you locate materials owned or licensed by your library. These include print books, ebooks, streaming videos, government documents, and other library resources.

The Open Web

Search engines such as Google are useful for finding background information, recent events, organizational websites, and search terms that can help you refine your research. Because anyone can publish online, the quality of information varies widely. Always evaluate web sources carefully before using them in your research.

Government and Statistical Sources

Government websites provide authoritative information on laws, public policies, official statistics, and government reports. They are often excellent sources for primary documents and reliable data.

Examples include the U.S. **Census Bureau**, the **Bureau of Labor Statistics**, the **Centers for Disease Control and Prevention (CDC)**, and state or local government websites.

News and Media

News organizations, podcasts, documentary films, television broadcasts, and social media provide timely information about current events and public discussions. Because reporting and commentary can vary in quality and perspective, researchers should compare information across multiple credible sources.

Research Tip

When using Google to find information from trusted organizations, you can narrow your results by adding a domain limit to your search.

- **site:gov** searches only U.S. government websites.
- **site:edu** searches only colleges and universities.
- **site:org** searches nonprofit organizations, although these sites should still be evaluated for authority and bias.

Using these search operators helps reduce irrelevant results and makes it easier to locate credible information.

Choosing the Right Tool for the Question

One of the keys to successful research is matching your question to the right search tool. The following guidelines can help you get started:

- **Need a clear explanation of an unfamiliar topic?** Start with a **tertiary source**, such as an encyclopedia or topic overview, to build background knowledge. Then move to scholarly sources for more in-depth information.
- **Need current data or statistics?** Government agencies and official statistical sources are usually the best place to find reliable, up-to-date information.
- **Need to know what is happening now?** Search newspapers and credible news websites for timely reporting and developing events.
- **Need evidence to support an academic argument?** Search library databases for peer-reviewed scholarly articles and other academic sources relevant to your topic.
- **Need an original document or firsthand evidence?** Look for primary sources, such as laws, speeches, interviews, historical documents, original research studies, photographs, or datasets. These can often be found through government websites, library archives, special collections, or library databases.

Going Further: The Deep Web

Much of the information available through libraries exists in what is called the **deep web**. Unlike the open web, the deep web includes content stored in library databases, catalogs, archives, and other resources that search engines such as Google cannot fully index.

As a result, a Google search may not retrieve many of the highest-quality sources available through your library. This is one reason academic research relies on library search tools in addition to general search engines.

Research Tip

When using Google to find information from trusted organizations, you can narrow your results by adding a domain limit to your search.

- **site:gov** searches only U.S. government websites.
- **site:edu** searches only colleges and universities.
- **site:org** searches nonprofit organizations, although these sites should still be evaluated for authority and bias.

Using these search operators helps reduce irrelevant results and makes it easier to locate credible information.

Conclusion

Chapter Summary

In this chapter, you learned that information follows recognizable patterns. The **information cycle** explains how different types of sources become available over time, helping you choose sources that match your topic and research question.

You also explored the **six types of information**: objective, subjective, opinion, propaganda, satire, and parody. Understanding the purpose of each type helps you evaluate sources and determine how they should be used in research.

Next, you examined three important distinctions: **objective versus subjective**, **scholarly versus popular**, and **primary, secondary, and tertiary** sources. These categories describe different characteristics of information and help you select evidence that is appropriate for your assignment.

Finally, you learned that effective research begins by choosing the right search tool. Library databases, library catalogs, government websites, news sources, and the open web each provide different kinds of information. Knowing where to search can save time and lead to stronger, more reliable research.

Key Takeaway

Successful researchers do more than find information. They understand what type of information they need, where to find it, and how to evaluate whether it is appropriate for their research question.

Key Concepts and Vocabulary

- **Information cycle.** The predictable pattern by which information about an event becomes available over time, from same-day social media to books years later.
- **Objective vs. subjective.** Factual information relayed without bias versus information that expresses analysis or opinion.
- **Propaganda.** False or selectively chosen information used to advance a political, ideological, or religious doctrine.
- **Scholarly vs. popular.** Peer-reviewed journals written by experts versus magazines and news written for general audiences.
- **Primary, secondary, and tertiary sources.** Firsthand evidence; analysis of that evidence after the fact; and consolidated overviews such as encyclopedias.
- **Library database.** A searchable collection of published sources, the best starting place for full-text scholarly articles.

Critical Thinking Exercises

Try It: Reflection Activities

Activity 2.1A — Map a Topic Through the Information Cycle

Choose a recent event (within the last year) and a historical event (more than 20 years ago). For each, identify what kinds of sources currently exist at each stage of the information cycle: day of, week of, week after, months after, and a year or more later. Where are the gaps for the recent event, and how should that shape your search strategy?

Activity 2.1B — Sort Twelve Sources

Choose a research topic and compare the results from Google with the results from two library databases. Which search finds more scholarly sources? What kinds of information appear in one search but not the other? Reflect on how your search strategy influences the evidence you discover.

Think About It: Discussion Questions

1. If you had to write a research paper on an event that happened last week, which stages of the information cycle would not yet be available to you? How would you adapt your strategy?
2. Have you ever been fooled by satire or parody you mistook for real news? What signal would have warned you?
3. Why do scholars insist that researchers use peer-reviewed sources, even though they take longer to publish than news or magazine articles?

Attributions and References

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2.2: Exploratory Research

Unit 2 Exploring the Information Landscape | Chapter 2.2

Learning Objectives

- Explain the purpose and value of exploratory research.
- Use background reference sources to build foundational knowledge about an unfamiliar topic.
- Evaluate AI-generated search summaries as exploratory tools.
- Use Wikipedia responsibly during exploratory research.
- Apply strategic Google searching techniques to learn about a topic.
- Recognize when expert guidance can improve your research.

Introduction

Every research project begins with uncertainty. At the start, you may know only a general area of interest or a broad question, but not enough to develop a clear research focus. Exploratory research is the process of learning about a topic before trying to answer a specific question. It helps you build background knowledge, understand key concepts, identify important debates, and discover the vocabulary experts use to discuss the subject.

This stage of research is not about collecting evidence for your paper. It is about becoming informed enough to ask better questions. The stronger your understanding of a topic, the easier it becomes to identify worthwhile research questions, locate relevant sources, and recognize credible information. Exploratory research lays the foundation for every successful research project.

Why Exploratory Research Comes First

Don't pick a topic. Pick a question you don't yet know the answer to.

Exploratory research is how you move from a vague interest to an informed understanding of a topic. It is the first step in investigating something new or unfamiliar. During this stage, you gather background information, identify important issues, learn the language experts use, and begin to understand the conversations surrounding a topic.

Think of exploratory research as reconnaissance. Before climbers scale a mountain, they study maps, examine the terrain, and identify possible routes. Before researchers begin searching for scholarly articles, they first learn enough about a topic to know what questions are worth asking.

Students who spend time exploring a topic first usually identify better search terms, locate more relevant sources, recognize important debates, and ultimately write stronger papers. Exploratory research is not about collecting evidence for your final paper. It is about learning enough to ask better questions later.

Don't Skip Exploratory Research

Skipping exploratory research is one of the most common reasons research projects go off the rails. Students who jump from "I want to write about social media" directly into a database search often retrieve hundreds or even thousands of articles without knowing which ones are relevant. The result is often a paper that wanders because the writer never developed a clear understanding of the topic.

What Exploratory Research Is For

Exploratory research serves four important purposes.

- **Clarify the Problem:** Many research topics begin as broad interests rather than clearly defined questions. Exploratory research helps you understand what the real issues are so you can decide what deserves further investigation.
- **Generate Ideas:** As you read, you discover new perspectives, related issues, and questions you may never have considered. These discoveries often become the foundation of a stronger research project.

- **Understand the Context:** Facts rarely exist in isolation. Exploratory research provides the historical, scientific, political, cultural, or social background needed to understand what later sources are discussing.
- **Build Your Vocabulary:** Every field has its own language. During exploratory research, you begin learning the keywords, technical terms, acronyms, and concepts that experts use. These terms later become valuable search keywords when you begin searching scholarly databases.

Tools for Exploratory Research

The goal of exploratory research is not to find sources you'll eventually cite. Instead, the goal is to become knowledgeable enough about your topic that you know what to search for, what questions to ask, and which sources deserve your attention. Different tools serve different purposes during this stage.

Background Reference Sources

Subject encyclopedias and topic overview databases are often the fastest way to become familiar with an unfamiliar topic.

Currently enrolled IVC students have 24/7 access to a variety of reference sources using the IVC credentials.

- *Britannica*
- *CQ Researcher*
- *Gale Virtual Reference eBooks* (subject encyclopedias)
- *Opposing Viewpoints*

Unlike scholarly journal articles, these resources summarize existing knowledge rather than present original research. They explain major concepts, introduce key terminology, identify influential researchers and organizations, and provide historical context.

IVC Database Access

To access the IVC Library databases, visit the [IVC Library's list of Databases A-Z](#). Use your IVC login credentials when prompted.

AI-Generated Search Summaries

Many search engines now generate AI-written summaries at the top of search results. These summaries can be useful for quickly becoming familiar with a topic, but they should never be treated as authoritative sources.

Unlike an encyclopedia article, an AI-generated summary does not have a stable author, editor, or publication process. It is created on demand by combining information from multiple websites, some of which may be inaccurate, outdated, biased, or taken out of context. Although these summaries often sound confident and well written, they can contain factual errors, omit important context, or combine information from different sources in misleading ways.

Use AI-generated summaries as conversation starters rather than final answers. They can help you identify unfamiliar concepts, discover useful keywords, and generate questions worth investigating. Once you have that starting point, verify what you learned using reference sources, scholarly literature, government publications, or other credible evidence.

AI Hallucinations

AI systems sometimes generate information that is false, fabricated, or unsupported by evidence. This phenomenon is known as an **AI hallucination**.

Imagine asking AI about a medical condition you've never heard of, a Supreme Court case you have never studied, or a historical event that is unfamiliar to you. If the explanation contains an error, would you recognize it? Probably not. Experts can often identify hallucinations because they already understand the subject. Beginners usually cannot.

This is why information literacy emphasizes verification instead of trust. Whenever AI introduces a new fact, statistic, quotation, or claim, confirm it using reliable sources before accepting it as accurate. Confidence is not evidence.

Using Wikipedia Responsibly

Students are often told never to use Wikipedia. That advice is too simplistic. Although Wikipedia is not an appropriate source for most college research papers because it is a tertiary source that anyone can edit, it is one of the best tools available for exploratory

research.

Many students assume AI-generated summaries are more reliable because they sound polished and authoritative. In reality, Wikipedia is often the better starting point because its evidence is transparent. Most factual claims are supported by citations, and readers can examine the references, edit history, and discussion pages to evaluate the information. AI-generated summaries typically do not reveal exactly which sources they used or how they reached their conclusions.

Neither Wikipedia nor AI-generated summaries should normally be cited in college research papers. Instead, use Wikipedia to:

- gain a quick overview of an unfamiliar topic,
- learn important vocabulary and concepts, and
- identify credible sources through the article's references and external links.

Those references often lead to books, scholarly articles, government reports, and other high-quality sources that you can locate through your library.

Strategic Google Searching

Google is useful during exploratory research as long as you treat its results as starting points rather than final answers.

Several simple search operators can dramatically improve the quality of your results. Adding any of the operators below to your search terms in the Google search box will help the algorithm produce more meaningful results.

- **site:gov** limits results to government websites.
- **site:edu** limits results to colleges and universities.
- **filetype:pdf** often uncovers reports, white papers, policy documents, and technical publications.
- Placing **quotation marks** around a phrase searches for that exact wording

These techniques reduce irrelevant results and help you find higher-quality information more quickly.

Searching Trusted Organizations

Another way to improve a Google search for exploratory information is to include the name of a reputable organization that specializes in your topic. These organizations often publish reports, statistics, policy briefs, and background information that are freely available online. Adding an organization's name to your search can help you find higher-quality information more quickly.

Examples include:

- **Pew Research Center** (public opinion and social trends)
- **RAND Corporation** (public policy and national security)
- **Brookings Institution** (economics and public policy)
- **Council on Foreign Relations (CFR)** (international affairs)
- **Center for American Progress** (public policy)
- **Cato Institute** (public policy)

Keep in mind that many organizations have specific missions or policy perspectives. Consider the organization's purpose and compare its information with other credible sources as you evaluate the evidence.

Remember that Google searches the open web. As your research becomes more focused, your library's databases should become your primary research tools because they provide access to scholarly literature that Google may not index or make freely available.

Developing Search Terms for Exploratory Research

To explore a topic effectively, you first need to develop a list of **keywords** (also called **search terms**). Keywords are the words and phrases you enter into search engines, library databases, and library catalogs to find information. Rather than searching with full questions, experienced researchers begin by identifying the important concepts and the vocabulary used to describe them.

Every topic has its own language. Researchers, journalists, policymakers, and practitioners often use different terms to describe the same idea. The more keywords you know, the more likely you are to discover relevant information.

Suppose you are interested in researching **artificial intelligence** and **intellectual property**. Searching with only those two keywords will produce useful information, but it will also miss sources that use more specific terminology.

✓ Example 2.2.1

As you begin exploratory research, your keyword list grows.

- **Artificial intelligence:** AI, generative AI, machine learning, large language models (LLMs), ChatGPT, artificial neural networks
- **Intellectual property:** copyright, patent, trademark, fair use, licensing, ownership, authorship
- **AI-generated content:** synthetic media, AI-generated art, AI-generated writing, computer-generated works
- **Legal concepts:** copyright infringement, copyrightability, copyright law, legal precedent, U.S. Copyright Office
- **Creative industries:** artists, authors, musicians, graphic designers, content creators, publishers

One of the easiest ways to discover new keywords is to ask yourself a few simple questions:

- What are other names for this topic?
- What broader or narrower terms describe it?
- What organizations, people, companies, or places are associated with it?
- What technical or legal terms appear repeatedly?
- Which keywords appear in article titles, abstracts, or subject headings?

📌 Research Tip

As you conduct exploratory research, keep a running list of useful keywords and phrases.

Record unfamiliar terms, important organizations, recurring concepts, and names that appear repeatedly. This growing keyword list becomes increasingly valuable because it improves every future search in library databases, search engines, and government websites.

✓ Example 2.2.2

As you continue reading, you will discover even more specialized keywords.

- U.S. Copyright Office
- World Intellectual Property Organization (WIPO)
- Copyright Alliance
- Authors Guild
- Training Data
- Fair Use Doctrine

Notice how your searches become more precise as your vocabulary grows. Instead of repeatedly searching **AI** and **intellectual property**, you begin asking more focused questions, such as:

- Who owns AI-generated artwork?
- Can AI-generated content be copyrighted?
- Does training AI on copyrighted material qualify as fair use?
- What guidance has the U.S. Copyright Office issued about AI-generated works?

Developing keywords is an ongoing process. Every source you read introduces new vocabulary that leads to better searches, more relevant sources, and a stronger research question. Learning the language of your topic is one of the most important skills in the research process.

Ask a Professional

One of the most valuable research tools is another person.

Reference librarians, professors, and professionals in the field can recommend key resources, important researchers, useful keywords, and effective search strategies. A brief conversation can save hours of searching and help you avoid pursuing an unworkable research idea.

Experienced researchers routinely seek advice from colleagues. Developing the same habit will make your research faster, more focused, and more effective.

✓ Ask a Librarian

IVC Librarians are here to help!

We offer in person and online research assistance, including citation assistance. [Find our hours and more on the IVC Library website.](#)

Conclusion

Exploratory research is the stage where curiosity becomes understanding. By building background knowledge, learning the vocabulary of a topic, and identifying its key issues, you create a strong foundation for every step that follows. Investing time in exploration makes it easier to ask better questions, search more strategically, and evaluate sources with confidence.

Going Further

Professional researchers rarely begin a project by immediately searching for scholarly articles. Instead, they often conduct **scoping searches**, a form of exploratory research that helps them understand the size, scope, and existing knowledge within a field.

In published research, exploratory work becomes more formal. Systematic reviews document their scoping searches, and grant proposals begin with background sections that summarize existing knowledge before proposing new research.

Although your exploratory research will probably be less formal, the purpose is the same. Before asking a meaningful question, you first need to understand the conversation.

Chapter Summary

Exploratory research is the foundation of every successful research project. Before searching for scholarly evidence, researchers first build background knowledge, learn the language of the field, and identify the major issues surrounding a topic.

Background reference sources, AI-generated search summaries, Wikipedia, strategic Google searching, and conversations with experts all support exploratory research, but each should be used differently. AI summaries can provide quick orientation, but they may also produce convincing hallucinations that are difficult for beginners to recognize. Wikipedia's transparent citations and editorial process often make it a more reliable starting point because readers can inspect the evidence behind its claims.

The goal of exploratory research is not to gather evidence for your paper. It is to become informed enough to recognize worthwhile questions, evaluate information critically, and begin your research with confidence.

Key Concepts and Vocabulary

- **Exploratory research.** First-stage gathering of background information that turns a vague interest into a researchable question.
- **Topic, subject, research question.** Three stages of focus, from broad area of interest to narrower angle to specific, answerable question.
- **Background reference source.** Subject encyclopedias and topic-overview databases used to get oriented before committing to a question.
- **Topic sustainability test.** Five checks (sources, scope, currency, personal investment, assignment fit) run before committing to a topic.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 2.2A — The Six-Question Background Map

Choose a topic that interests you. Spend 30–45 minutes with a topic encyclopedia or current-issues database (Britannica, CQ Researcher, Gale Opposing Viewpoints, or similar). Write a one-paragraph answer to each of the six background questions. By

the end, you'll have a 600–900 word "map" of the topic that will save you hours later.

Activity 2.2B — Wikipedia Reference Mining

Pick a Wikipedia article on your topic. Don't quote from it. Instead, scroll to the References section at the bottom and select three citations that look credible — peer-reviewed articles, books from university presses, government reports. Find each one through your library, read at least the abstract, and write a 100-word note on what each source contributes.

? Think About It: Discussion Questions

1. Most students try to skip exploratory research. Why do you think that is? What makes the temptation to skip it so strong?
2. Wikipedia is treated as untouchable in some classrooms and as a key tool in others. How do you reconcile those two stances?

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2.3: Refining a Research Topic

Unit 2 Exploring the Information Landscape | Chapter 2.3

Learning Objectives

- Differentiate between a broad topic and a focused subject.
- Apply six background questions to refine a broad topic into a manageable area of study.
- Recognize scholarly debates by identifying areas of disagreement, unanswered questions, and emerging issues.
- Evaluate the sustainability of a research topic based on scope, available evidence, currency, personal interest, and assignment requirements.
- Revise the scope of a topic by narrowing or broadening it before developing a research question.

Introduction

Exploring a topic is only the first step in the research process. Once you have built background knowledge, you need to narrow your focus to a topic that is specific enough to investigate but broad enough to support meaningful research. This refinement is what transforms a general interest into a manageable research project.

In this chapter, you'll learn how to move from a broad topic to a focused subject by asking strategic background questions, recognizing promising areas for investigation, and evaluating whether a topic is sustainable for a semester-long project. Every successful research project begins with a broad topic, narrows to a focused subject, and then develops a research question. This chapter focuses on the middle step: refining the topic.

Topic, Subject, and Scope

The terms *topic* and *subject* are often used interchangeably, but they represent different levels of focus in the research process. Understanding the difference will help you narrow a broad idea into one that is manageable for a research project.

A **topic** is the broad area you want to explore. It identifies the general subject matter but is too broad to guide research on its own. For example, **artificial intelligence** is a topic.

A **subject** is a narrower aspect of that topic. It focuses on a specific issue, population, place, or perspective and provides a manageable direction for further investigation. For example, **the use of artificial intelligence in college writing courses** is a subject.

The difference between a topic and a subject is largely a matter of **scope**. Scope refers to how broad or narrow a research project is. A topic with a broad scope covers many issues and can quickly become overwhelming. A topic with a narrow scope focuses on a smaller area that can be explored in greater depth. The goal is to find a scope that is appropriate for the assignment: narrow enough to investigate thoroughly, but broad enough to support meaningful research.

Many beginning researchers try to move directly from a broad topic to a research question. Refining the topic into a focused subject is the essential step in between. It helps you understand what scholars have already studied, identify promising areas for further investigation, and determine whether your topic is manageable. In the next chapter, you'll use that refined subject to develop a focused research question.

6 Questions to Refine Your Topic

Once you've chosen a topic and begun your exploratory research, the next step is to map the territory. Before you can refine a topic into a focused subject, you need to understand it from multiple perspectives.

The following six questions provide a framework for building that background knowledge and can be applied to almost any research topic. As you answer these questions, you'll often find that your topic's scope begins to change. Each answer helps narrow a broad idea into a more focused subject.

1. How did the topic emerge?

When did the topic become a subject of public or scholarly interest? Consider the events, technological developments, social changes, or discoveries that brought attention to it.

2. What laws or policies are involved?

Identify the laws, regulations, institutional policies, or professional guidelines related to the topic. Are there policy debates or gaps that researchers frequently discuss?

3. Who are the key actors?

Determine which individuals, organizations, government agencies, corporations, advocacy groups, or professional associations play important roles in the issue.

4. What vocabulary do experts use?

Learn the specialized terminology used by researchers and professionals. Understanding the language of the field will improve both your comprehension and your database searches.

5. What problems or unanswered questions exist?

Identify the major challenges, controversies, limitations, or unresolved questions surrounding the topic. These often point toward potential research questions.

6. What benefits or opportunities exist?

Consider the positive outcomes, successful approaches, innovations, or opportunities associated with the topic. Examining both strengths and challenges leads to a more balanced understanding.

Working through these six questions before settling on a research question helps you build the background knowledge needed for effective research. Instead of choosing a question based on first impressions or personal opinions, you will be better prepared to refine your topic into a focused subject and eventually develop a researchable question.

Following Scholarly Conversations

As you read background sources, pay attention to the places where the conversation becomes more complex. Promising research topics often emerge where scholars identify uncertainty, disagreement, or unanswered questions.

Look for these three signals:

- **Where do experts disagree?** Differences in interpretation, competing explanations, or conflicting recommendations often point to meaningful research questions.
- **Where is the evidence limited?** Topics that receive significant public attention but have relatively little scholarly research can provide opportunities for further investigation, especially in undergraduate research.
- **Where are the ethical, social, or political tensions?** Issues involving competing values, public policy, or difficult tradeoffs often generate sustained scholarly discussion and require researchers to consider their own positionality (Chapter 1.3).

Where Strong Topics Emerge

The strongest research questions rarely come from ideas that everyone already accepts. Instead, they emerge from the edges of the conversation, where scholars continue to debate interpretations, where evidence remains incomplete, or where new developments challenge existing knowledge. As you conduct exploratory research, learn to recognize these moments. They often provide the foundation for questions that are both engaging and worth investigating.

The Topic Sustainability Test

Before investing significant time in a topic, make sure it can support a successful research project. If you cannot answer **yes** to each one, your topic may need to be revised.

1. Are enough credible sources available?

Can you find sufficient scholarly and other reliable full-text sources to support your research?

2. Is the scope appropriate?

Is the topic narrow enough to investigate thoroughly but broad enough to support meaningful analysis?

3. Is the topic current?

Is there enough recent scholarship to support your research, or is the available literature outdated?

4. Are you genuinely interested in the topic?

Will you remain curious and motivated as you read, analyze, and write about it over several weeks?

5. Does the topic fit the assignment?

Does it address the goals, requirements, and expectations of your assignment and course?

If the Answer Is No

A topic that fails one of these tests does not have to be abandoned. In many cases, you can improve it by narrowing or broadening the scope, changing your perspective, or connecting it to a related issue with stronger evidence. If you're unsure how to revise your topic, ask a librarian for help before starting over.

Ask a Professional

One of the most valuable research tools is another person.

Reference librarians, professors, and professionals in the field can recommend key resources, important researchers, useful keywords, and effective search strategies. A brief conversation can save hours of searching and help you avoid pursuing an unworkable research idea.

Experienced researchers routinely seek advice from colleagues. Developing the same habit will make your research faster, more focused, and more effective.

✓ Ask a Librarian

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Conclusion

Refining a research topic is the bridge between curiosity and investigation. By narrowing a broad area of interest into a focused, manageable subject, you create a foundation for effective searching, thoughtful analysis, and meaningful research. A well-defined topic saves time, improves the quality of your sources, and keeps your project focused from beginning to end.

The next chapter builds on this foundation by showing how to transform a refined topic into a clear, focused, and researchable question that will guide the rest of your project.

Chapter Summary

Refining a research topic transforms a broad area of interest into a focused subject that can support meaningful research. Using strategic background questions helps you better understand a topic, identify key issues, recognize important organizations and terminology, and discover opportunities for further investigation. Paying attention to scholarly debates, gaps in evidence, and emerging issues can reveal promising directions for research. Before moving forward, a topic should also be evaluated for its scope, available evidence, currency, personal interest, and fit with the assignment. A well-refined topic provides the foundation for the next step: developing a focused research question that will guide the rest of your project.

Key Concepts & Vocabulary

Topic. A broad subject or area of interest that serves as the starting point for research.

Subject. A narrower aspect or perspective within a broad topic that provides a more manageable focus for investigation.

Scope. The breadth or narrowness of a research topic. A topic with an appropriate scope is focused enough to investigate thoroughly but broad enough to support meaningful research.

Topic refinement. The process of narrowing a broad topic into a focused, researchable subject through exploratory reading and critical thinking.

Background questions. A set of guiding questions used to better understand a topic by examining its history, key actors, terminology, current issues, laws and policies, and potential benefits.

Scholarly debate. An area of disagreement, competing interpretations, or unresolved questions within the scholarly literature that may lead to productive research.

Topic sustainability. The likelihood that a topic can support a successful research project based on its scope, available evidence, currency, personal interest, and alignment with the assignment.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 2.3A — Refine a Broad Topic

Choose a broad topic that interests you, such as artificial intelligence, climate change, mental health, or social media. Refine it into a focused subject by answering the six background questions presented in this chapter. Then write one paragraph explaining how your topic became more focused and what you learned through the process.

Activity 2.3B — Test Your Topic

Evaluate your proposed research topic using the Topic Sustainability Test. For each of the five criteria (source availability, scope, currency, personal interest, and assignment fit), explain whether your topic passes the test and why. If your topic falls short in any area, describe how you would narrow, broaden, or otherwise revise it to make it more suitable for a semester-long research project.

? Think About It: Discussion Questions

1. Why is it important to narrow a broad topic before beginning a research project?
2. Why are scholarly debates good places to begin research?

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Badke, W. (2021). *Research strategies: Finding your way through the information fog* (7th ed.). iUniverse

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2.4: Crafting a Sustainable Research Question

Unit 2 Exploring the Information Landscape | Chapter 2.4

Learning Objectives

- Distinguish among a topic, a claim, and a research question.
- Evaluate a research question using four criteria for quality.
- Use question-development frameworks such as PICO and the 5Ws + H to generate research questions.
- Revise a research question by refining its scope and focus.

Introduction

Every research project begins with a question. Before searching for evidence, evaluating sources, or developing an argument, researchers must decide exactly what they want to investigate. A strong research question provides direction, establishes the scope of the project, and guides every decision that follows. A weak question, however, can lead to unfocused searching, irrelevant sources, and superficial conclusions.

In the previous chapters, you explored a topic, built background knowledge, and identified scholarly conversations surrounding your subject. Now it is time to transform that broad understanding into a clear, focused, and researchable question. In this chapter, you'll learn how to distinguish a topic from a research question, evaluate the quality of your question, use proven frameworks to generate ideas, and refine your question through multiple revisions. By the end of the chapter, you'll have a research question that can support a meaningful and sustainable research project.

From Topic to Question

A strong research question is focused enough to investigate, important enough to matter, and open enough that the answer is not already known. Chapter 2.2 introduced the exploratory research process, where you build background knowledge and learn what scholars already know about a topic. This chapter explains how to use that knowledge to develop a clear, researchable question.

One of the most common mistakes new researchers make is confusing a **topic**, a **claim**, and a **research question**. *Social media* is a topic. *Social media is bad for teenagers* is a claim or conclusion. Neither is a research question.

A research question asks something that can be investigated with evidence. It does not assume the answer in advance. Instead, it invites inquiry, allowing the evidence to shape the conclusion. A well-designed research question is clear enough that another researcher could investigate the same question and evaluate the evidence independently.

Four Tests for a Sustainable Research Question

Once you've narrowed your topic, evaluate your research question before committing to it. A strong question should pass all four of these checks.

1. Does it align with the assignment?

Your research question should match the purpose, scope, and expectations of the assignment. Even an excellent question is ineffective if it does not address the task you were given.

2. Is it open-ended?

A research question should invite investigation rather than a simple yes-or-no answer. Questions that begin with **how**, **why**, **what**, or **to what extent** usually encourage deeper analysis than questions beginning with **does** or **will**.

3. Is there enough credible evidence available?

Before committing to a question, conduct a brief exploratory search in a library database. If you cannot find enough reliable, full-text sources, the question may be too narrow or focus on an area that has not been sufficiently studied.

4. Is the level of complexity appropriate?

The question should be focused enough to answer within the scope of the assignment while still allowing for meaningful analysis. If it is too broad, your research will become unfocused. If it is too narrow, you may struggle to find enough evidence or develop a substantial discussion.

Crafting a Strong Research Question

The strongest research questions do not ask you to prove what you already believe. They ask you to investigate what the evidence reveals.

- **They ask a question instead of making a claim.** A research project begins with curiosity, not a conclusion.
- **They use neutral language.** Strong questions avoid emotionally charged words such as *destroying*, *bad*, or *invasive*, allowing the evidence to shape the findings.
- **They are open-ended.** Rather than producing a simple yes-or-no answer, they invite explanation, analysis, and interpretation.
- **They are clear and focused.** The question is specific enough to investigate within the scope of the assignment while remaining broad enough to support meaningful analysis.

One of the easiest ways to recognize a strong research question is to compare it with weaker versions. The examples below show how a question improves as it becomes more focused, neutral, and researchable.

✓ Example 2.4.1

Deepfakes and Election Information

Weak: *Deepfakes are destroying democracy.*

This is a claim, not a research question.

Better: *Why are deepfakes destroying democracy?*

This is a question, but it assumes the conclusion before the research begins.

Strong: *How are AI-generated deepfakes influencing public trust in election information?*

This question is open-ended, uses neutral language, and invites investigation rather than assuming a particular outcome.

✓ Example 2.4.2

Social Media and Politics

Weak: *Does social media use cause bad thinking and political problems in the United States?*

This is a yes-or-no question with emotionally charged language.

Better: *How does the use of social media negatively impact people's opinions and contribute to political polarization in the United States?*

The question is open-ended, but it still assumes negative effects and is unnecessarily wordy.

Strong: *How has social media influenced public opinion and political polarization in the United States?*

This version is concise, neutral, and allows the evidence to determine the relationship between social media and political polarization.

✓ Example 2.4.3

Data Mining and Consumer Privacy

Weak: *How does data mining for targeted marketing create risks that consumers' personal information will be leaked and misused?*

This question is overly specific and presumes particular outcomes.

Better: *Why are personally invasive algorithms so important in social media?*

This version is too broad, uses subjective language, and lacks a clear research focus.

Strong: *How does data mining for targeted marketing affect consumer privacy?*

This question is focused, neutral, and broad enough to support meaningful research.

Neutral wording does not mean researchers lack opinions. Instead, it means they postpone judgment until they have examined the evidence. Beginning with a neutral question reduces confirmation bias and encourages a fair evaluation of competing explanations. A research project should follow the evidence wherever it leads, even if the findings differ from the researcher's initial expectations.

Question Frameworks That Help

If you're struggling to turn a topic into a research question, a structured framework can provide a useful starting point. These tools help generate ideas, identify possible directions, and organize your thinking. Two frameworks are especially useful for introductory research projects.

PICO

Originally developed for health care research, the PICO framework can be adapted to many research questions by breaking a topic into its key concepts. PICO stands for:

- **Population** – Who or what is being studied?
- **Intervention** (or issue, condition, or exposure) – What is being examined?
- **Comparison** – What is the intervention being compared with, if anything?
- **Outcome** – What effect or result is being measured?

For example:

Among high school students (Population), how does nighttime smartphone use (Intervention), compared with no nighttime smartphone use (Comparison), affect sleep duration (Outcome)?

The 5Ws + H

Another useful approach is to ask the classic journalistic questions: **Who? What? When? Where? Why?** and **How?** Each answer suggests a different angle for investigating a topic and can lead to a distinct research question. This framework works particularly well during exploratory research because it encourages you to consider multiple perspectives before settling on a single question.

For example, suppose your topic is **food insecurity among college students**.

- **Who?** Community college students
- **What?** Food insecurity
- **Where?** Southern California
- **When?** During the academic year
- **Why?** Academic performance
- **How?** Campus support programs

One research question generated from these responses might be:

How does food insecurity affect the academic performance of community college students in Southern California?

Remember that these frameworks are tools, not formulas. You do not need to force every topic into PICO or the 5Ws + H. Instead, use them as scaffolds to generate possible research questions, then refine the one that best fits your topic, assignment, and the available evidence.

Revising Your Question (Iteration)

Your first research question is rarely your final one. As you learn more about your topic, you'll often discover that the question needs to be refined. Plan to revise it at least once or twice, and expect it to continue evolving as your research progresses.

Two of the most common revisions involve adjusting the scope.

Narrow the question if it is too broad. Signs include finding thousands of relevant sources, struggling to decide what to include, or realizing the question covers too many issues. You can narrow the scope by focusing on a particular population, location, time period, or aspect of the topic.

Broaden the question if it is too narrow. If you find very few relevant sources or the available evidence does not address your question, consider expanding the population, widening the time frame, or shifting from a specific case to a broader issue.

When the Question Changes

It is normal, and often beneficial, for your research question to change as you continue reading. New evidence may reveal that your original question is too broad, too narrow, no longer relevant, or simply less interesting than another direction suggested by the literature. Revising your question is not a sign that your project is off track. Instead, it shows that you are responding to the evidence and allowing your understanding of the topic to develop. Experienced researchers routinely revise their questions as they encounter new evidence. The goal is not to defend the original question but to ask the best question the available evidence can answer.

By the time you begin writing, your research question should accurately reflect the investigation your paper actually conducts.

Ask a Professional

One of the most valuable research tools is another person.

Reference librarians, professors, and professionals in the field can recommend key resources, important researchers, useful keywords, and effective search strategies. A brief conversation can save hours of searching and help you avoid pursuing an unworkable research idea.

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✓ Ask a Librarian

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Conclusion

A well-crafted research question is the foundation of every successful research project. It transforms a broad topic into a focused investigation, guides your searches for evidence, and helps you stay focused as your understanding of the topic develops. Strong research questions are clear, neutral, appropriately scoped, and supported by enough credible evidence to sustain meaningful inquiry. Although your question will almost certainly evolve as you continue researching, that revision is a normal part of the scholarly process. Good researchers refine their questions in response to what the evidence reveals.

Going Further: From Research Question to Research Proposal

In many upper-division courses, honors programs, and undergraduate research conferences, students move beyond developing a research question to writing a research proposal. A proposal explains the question being investigated, why it matters, what previous research has found, and how the study or investigation will be conducted. A clear, focused research question serves as the foundation for every part of the proposal. If you can explain your question, its significance, and your planned approach in a few concise paragraphs, you have built a strong foundation for the research that follows.

Chapter Summary

A research question transforms a broad topic into a focused investigation. Strong research questions align with the assignment, are open-ended, have sufficient credible evidence available, and define an appropriate scope for the project. Rather than beginning with a conclusion, effective researchers ask neutral questions that allow the evidence to shape their findings. Frameworks such as PICO and the 5Ws + H provide useful starting points for generating and refining research questions, while exploratory searching helps determine whether enough scholarly evidence exists to support the investigation. Because research is an iterative process, most questions are revised several times before reaching their final form. Refining a research question is not a sign that something is wrong; it is a hallmark of thoughtful, evidence-based inquiry.

Key Concepts & Vocabulary

Research question. A real, open question whose answer you do not yet have, framed so that others could investigate the same thing.

Open-ended question. A question that cannot be answered yes or no, usually opening with how or why.

PICO. A question framework identifying Population, Intervention, Comparison, and Outcome.

5Ws + H. Walking a topic through who, what, when, where, why, and how to generate candidate angles.

Narrowing and broadening. The two standard revisions of a research question, adjusting scope to match the available evidence.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 2.4A — Three Drafts of One Question

Take a topic from your background research (Chapter 2.2). Write three drafts of a research question: (1) a deliberately bad version (a conclusion or a yes/no question), (2) an okay version that still has problems, and (3) an excellent version that passes all four self-checks. Note in 100 words what you changed from one to the next.

Activity 2.4B — Database Pre-Search

Take your strongest candidate question and run a quick search in two library databases. Are there enough credible, full-text sources for a literature review? If not, what could you change about the question? Submit your search results and a 200-word reflection.

? Think About It: Discussion Questions

1. Look at the weak version research questions in the provided examples. Which kind of mistake do you find yourself making most often (turning a question into a conclusion, using negative framing, or being too wordy)?
2. Why do you think open-ended questions tend to produce stronger research papers than yes/no questions?

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CHAPTER OVERVIEW

3: Building and Refining Search Strategies

[3.1: Search Strategies and Database Skills](#)

[3.2: Research As an Iterative Process](#)

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3.1: Search Strategies and Database Skills

Unit 3 Building & Refining Search Strategies | Chapter 3.1

Learning Objectives

- Translate a research question into a strategic set of keywords.
- Use Boolean operators (AND, OR, NOT), phrase searching, and truncation to control your search.
- Distinguish between keywords and subject headings, and use both to your advantage.
- Apply the right filters (date, peer-review, source type) without over-narrowing your results.
- Document your search strategy so you can reproduce, share, or revise it.

Introduction

Searching is not just typing words into a search box and hoping for the best. Effective researchers approach searching as a strategic process. They begin with a clear research question, identify the most important concepts, choose appropriate keywords, and refine their searches based on what they discover.

Library databases are designed to help you find reliable scholarly sources, but they work differently from Google. The quality of your results depends largely on the quality of your search strategy. Learning a few simple techniques, such as using Boolean operators, phrase searching, truncation, subject headings, and filters, can dramatically improve both the relevance and efficiency of your searches.

In this chapter, you'll learn how to translate a research question into an effective search strategy, navigate library databases with confidence, and develop the search skills used by experienced researchers. By the end of the chapter, you'll spend less time sorting through irrelevant results and more time finding the evidence you need.

From Question to Keywords

Google rewards your first guess. Library databases reward your second, third, and fourth. Experienced researchers do not expect to find the perfect search on the first try. They expect to refine their searches as they learn more about their topic.

The first step is translating your research question into search terms. Start by identifying the two or three main concepts in your question. Then, for each concept, brainstorm synonyms, related terms, broader terms, narrower terms, and alternate spellings. The goal is to anticipate the language used by researchers, which may differ from the words you would naturally choose.

Building a Keyword Grid

Suppose your research question is: **How do urban green spaces affect air quality in large cities?** Break the question into its main concepts, then create a keyword bank for each one.

Building a Keyword Bank from a Research Question

Urban Green Spaces	Air Quality	Large Cities
parks	air pollution	urban areas
green spaces	particulate matter	metropolitan areas
urban forests	air quality	cities
street trees	emissions	municipalities
vegetation	atmospheric pollution	megacities

Once you have a keyword bank, you can mix and match terms to build different searches. For example, you might search:

- parks AND air pollution AND cities

- "urban forests" AND emissions AND metropolitan areas
- vegetation AND "air quality" AND urban areas

If one search produces poor results, substitute a different synonym from one or more columns and try again. Researchers rarely find the best search on their first attempt. Building a keyword grid gives you multiple ways to explore the same research question without starting over.

Keywords vs. Subject Headings

Library databases use two types of search terms: **keywords** and **subject headings**.

Keywords are the words you choose yourself. They include the terms, synonyms, and related ideas you brainstorm from your research question.

Subject headings are official terms assigned to each article by professional indexers. These terms come from a **controlled vocabulary**, meaning that every article about the same concept receives the same subject heading, even if the authors use different words.

This distinction matters because researchers often describe the same idea in different ways. One article might use the term *automobile*, while another uses *car* or *vehicle*. A keyword search must account for these variations. Subject headings solve this problem by grouping all articles on the same topic under a single standardized term.

Subject heading searches usually return fewer results than keyword searches because they retrieve only articles tagged with that specific topic. Fewer results do not necessarily mean a better search. If you find too few sources, switch back to keywords or combine both approaches.

Using Both Together

A good search strategy is flexible. Sometimes a keyword search is all you need. Other times, subject headings help you find articles that keywords alone might miss.

A common approach is to begin with a keyword search to identify several relevant articles. Then open one of the best results and look for the **Subject Headings** or **Descriptors** listed in the record. Those standardized terms can be used in a new search to find additional articles on the same topic.

You can also combine keywords and subject headings in a single search. This often makes your search more focused, but it can also reduce the number of results. If you retrieve too few articles, broaden your search by using additional keywords, removing a subject heading, or searching with keywords alone.

There is no single "right" way to search. Experienced researchers experiment with different combinations of keywords, subject headings, and filters, then evaluate the results. If the articles are relevant, your strategy is working. If they are not, revise your search and try again.

The Boolean Toolkit: AND, OR, NOT

Boolean operators are the words **AND**, **OR**, and **NOT** used to combine search terms. Although they sound technical, you already use them every day. In fact, most library databases, and Google, automatically insert **AND** between the words you type, so you've been performing Boolean searches all along.

These operators control the relationship between your search terms and determine which results are retrieved. Learning when to use each operator is one of the simplest and most effective ways to improve your searches.

AND narrows your search by requiring that **all** of the specified terms appear in each result. Use **AND** when you want to combine multiple concepts into a focused search. For example, searching **teenager AND social media AND mental health** retrieves only sources that discuss all three topics together.

OR broadens your search by retrieving results that contain **any** of the listed terms. It is most often used to connect synonyms, related terms, or alternative spellings so that you do not miss relevant sources. For example, **teenager OR adolescent OR youth** finds results that use any of those terms.

NOT excludes a specific term from your search results. This operator can be helpful when a keyword has multiple meanings, but it should be used sparingly because it may remove useful sources. For example, **pitbull NOT rapper** retrieves information about the dog breed while excluding results about the musical artist.

A Useful Way to Remember the Difference

Imagine you're at an ice cream shop.

- If you say, "I want **chocolate OR vanilla OR strawberry**," you're happy with any of those flavors. That is a **broad** search because it provides multiple options.
- If you say, "I want **chocolate AND vanilla AND strawberry**," you want all three flavors together. That is a **narrow** search because every condition must be met.
- Finally, if you say, "I want **chocolate AND vanilla NOT strawberry**," you're asking for chocolate and vanilla while excluding strawberry. Database searches work the same way.

Using NOT to Clarify Your Search

The **NOT** operator is most useful when a search term has multiple meanings. For example, searching **enterprise NOT "Star Trek"** retrieves information about the business concept rather than the fictional starship. Similarly, **groom NOT wedding** focuses on grooming instead of marriage-related topics. Because **NOT** removes results, use it carefully. It is most effective for eliminating an unwanted meaning of a word rather than broadly excluding information.

Search Modifiers

Boolean operators help you combine concepts, but they are only part of an effective search strategy. You can further improve your results by controlling **how** databases interpret your search terms. Two of the most useful techniques are **phrase searching**, which keeps words together in the exact order you specify, and **truncation**, which searches for multiple forms of the same word. Together, these techniques help you retrieve more relevant results while reducing unnecessary searching.

Phrase Searching with Quotation Marks

Databases search for your keywords wherever they appear in a document. If you search for **social media** without quotation marks, the database may retrieve articles in which *social* appears in one paragraph and *media* appears somewhere else, even though the phrase **social media** is never used. Placing quotation marks around a phrase tells the database to search for those words together, in that exact order.

Use quotation marks for established names, concepts, and expressions, such as:

- "social media"
- "climate change"
- "post-traumatic stress disorder"
- "electric current"
- "once in a blue moon"

Searching for phrases with quotation marks reduces irrelevant results and helps you find sources that discuss your topic more directly.

Truncation with the Asterisk

Truncation lets you search for multiple forms of a word at the same time. In most databases, the asterisk (*) is the truncation symbol. Placing it at the end of a word root tells the database to retrieve words that begin with that root but have different endings.

For example:

- **teen*** retrieves **teen, teens, teenager, teenagers, and teenage.**
- **child*** retrieves **child, children, and childhood.**
- **educat*** retrieves **educate, educator, educators, education, and educational.**
- **polit*** retrieves **politic, political, politics, politician, and politicians.**

Use truncation when several forms of a word are likely to appear in relevant sources. Choose a word root that is specific enough to avoid unrelated results. For example, **educat*** works well because it captures closely related terms without introducing much

irrelevant content. In contrast, very short roots such as **ed*** or **pol*** retrieve so many unrelated words that they usually create more noise than useful results.

Although the asterisk (*) is the most common truncation symbol, some databases use a different character. If truncation does not work as expected, consult the database's **Help** page to confirm its search syntax.

Using Search Filters

After you run a search, most library databases provide **filters** (sometimes called *limiters* or *facets*) that help you narrow your results without changing your search terms. Filters are especially useful when your initial search returns hundreds or thousands of sources.

The most commonly used filters include:

- **Publication date** narrows results to a specific time period. This is especially useful for rapidly changing topics such as artificial intelligence, public health, or climate policy.
- **Peer-reviewed (Scholarly) Journals** limits results to articles that have undergone peer review, a common requirement for college research assignments.
- **Full Text** displays only sources you can access immediately, saving time by excluding citations without available articles.
- **Source Type** lets you focus on specific kinds of publications, such as scholarly journals, newspapers, magazines, books, or trade publications.
- **Language** limits results to a particular language when a database includes multilingual content.

Filters are most effective **after** you have developed a strong search strategy using keywords, Boolean operators, phrase searching, and truncation. They refine a good search—they do not replace one.

Book Reviews

Watch for the book-review trap

When you filter for peer-reviewed articles, some databases will still return book reviews. Book reviews are not appropriate research evidence — they are short evaluations of a book, not original scholarship. You can usually spot them in the first line of the abstract or by a price listed in the record. Skip them and keep going.

Where Google Scholar Fits

Google Scholar sits between the open web and the library database. It searches scholarly literature with Google's convenience, its “Cited by” links power the forward citation chaining you will learn in Chapter 3.2, and it sometimes surfaces work that individual databases miss. Its interface requires no training, which is exactly why students reach for it first.

Use it with its limits in mind. Google Scholar has no controlled vocabulary, so there are no subject headings to sharpen a search. It offers few of the filters covered in this chapter, does not distinguish peer-reviewed journals from predatory ones, and often links to paywalled versions of articles your library provides for free. A practical workflow: explore in Google Scholar, then take the promising titles into your library's databases, where you can verify peer review and read the full text at no cost. Many libraries also offer a settings option that adds direct links from Google Scholar results to your library's full-text holdings.

Documenting Your Search Strategy

Strategic searchers keep a search log. The log doesn't need to be elaborate — a simple table works. For each search you run, record the database, the search terms, the filters you used, the number of results, and a brief note on whether it was useful.

A search log does three things. First, it keeps you from running the same dead-end search twice. Second, it lets you reproduce a successful search later, which matters when you need a few more sources weeks down the line. Third, it makes the methodology section of your literature review (Chapter 5.2) much easier to write — because you've already done the documentation.

Ask a Professional

One of the most valuable research tools is another person.

Reference librarians, professors, and professionals in the field can recommend key resources, important researchers, useful keywords, and effective search strategies. A brief conversation can save hours of searching and help you avoid pursuing an

unworkable research idea.

Experienced researchers routinely seek advice from colleagues. Developing the same habit will make your research faster, more focused, and more effective.

✓ Ask a Librarian

IVC Librarians are here to help!

We offer in person and online research assistance, including citation assistance. [Find our hours and more on the IVC Library website.](#)(opens in new window)

Conclusion

Effective searching is a skill that improves with practice. Every search teaches you something, whether it reveals useful sources, better terminology, or a new direction for your research. Instead of expecting the perfect search on your first attempt, think of searching as an iterative process of testing, evaluating, and refining. The strategies in this chapter provide a foundation for finding scholarly evidence efficiently and confidently, no matter what topic you investigate.

Going Further: Building Complex Search Strings

Professional researchers often combine multiple search techniques into a single, carefully constructed search string. For example: ("electric vehicles" OR EV OR "battery electric vehicle") AND (adoption OR adoption rates OR acceptance) AND (rural OR "rural communities") Searches like this use parentheses to group synonyms, Boolean operators to connect concepts, quotation marks to preserve phrases, and multiple keyword variations to capture the language used by different authors. In systematic reviews and other forms of rigorous research, these search strings are published in full so that other researchers can reproduce the search exactly. For an honors extension, build one nested search string for your own research question, run the search, and revise it at least twice. Record how each revision changes the number and relevance of your results.

Chapter Summary

Strategic searching transforms a research question into a systematic, reproducible process. Begin by identifying your main concepts and creating a keyword bank with synonyms and related terms. Use Boolean operators (AND, OR, and NOT) to combine concepts, quotation marks to search exact phrases, and truncation to capture multiple forms of a word. Combine keyword searches with subject heading searches to balance broad exploration with precise retrieval, then use filters to refine your results without unnecessarily excluding relevant sources. As you search, document what you tried so you can build on successful strategies and avoid repeating unsuccessful ones. The goal is not to find the perfect search on the first attempt, but to improve each search until you locate the evidence you need.

Key Concepts and Vocabulary

- **Boolean operators.** AND (narrows), OR (broadens), and NOT (excludes): the words that combine search terms.
- **Phrase searching.** Quotation marks that require words to appear together, in order, as in "social media".
- **Truncation.** An asterisk on a word root (teen*) that returns every ending of the word.
- **Keyword.** A natural-language search term you generate yourself.
- **Subject heading.** An official tag from a controlled vocabulary, assigned to every item on a concept regardless of the words its author used.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 3.1A — Build a Keyword Grid

Take your research question (from Chapter 2.3) and identify two or three main concepts. For each concept, brainstorm at least four synonyms or related terms. You should end up with a grid of 8–12 keywords. Use this grid to construct your search.

Activity 3.1B — Same Search, Three Databases

Run the same search in three different databases (e.g., ProQuest, Academic Search Complete, JSTOR, or whichever your library offers). Note (a) the number of results, (b) the kinds of sources each database returned, and (c) which database was strongest for your question. Submit a 200-word reflection.

? Think About It: Discussion Questions

1. Why do you think most students give up on a database search after one try and switch to Google? What would have to change for that habit to break?
2. "A search you didn't write down is a search you'll have to run again." Agree or disagree?

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3.2: Research As an Iterative Process

Unit 3 Building & Refining Search Strategies | Chapter 3.2

Introduction

Many students imagine research as a series of steps: choose a topic, search for sources, write the paper, and submit the assignment. Experienced researchers know the process is much more flexible. Every search reveals new information, every source raises new questions, and every new question changes the next search.

Research is an iterative process. Rather than moving in a straight line, researchers continually search, read, reflect, and revise their approach. Each cycle improves both their understanding of the topic and the quality of the evidence they collect. The goal is not to find the perfect source on the first attempt but to gradually build a deeper understanding of the scholarly conversation.

Research Is a Spiral, Not a Straight Line

Think of research as a spiral rather than a checklist. Each search teaches you something, whether it produces excellent results or very few at all. New terminology, unexpected perspectives, and recurring authors all help shape your next search.

This process is not a sign that you are doing research incorrectly. Revising your keywords, narrowing your focus, or even changing your research question reflects how experienced researchers work.

Common Mistake

Students often make the mistake of writing their paper first and then searching for evidence to support what they have already decided to say. This approach encourages confirmation rather than investigation. Good research allows the evidence to shape the argument, even when the findings challenge your original assumptions.

Most of your research time should be spent searching, reading, evaluating, and refining your understanding. By the time you begin writing, the research has already influenced what you have to say.

Joining the Scholarly Conversation

Every research project joins an ongoing scholarly conversation. Long before you began your project, researchers were asking questions, testing ideas, and debating evidence. Before contributing your own perspective, you first need to understand what others have already discovered.

Imagine joining a video meeting that has been underway for thirty minutes. You would probably spend a few minutes listening before speaking. You would want to understand the topic, recognize the different viewpoints, and avoid repeating points that have already been made.

Research works the same way. The articles and books you read early in the process help you understand the major ideas, identify areas of agreement and disagreement, and recognize questions that still need answers. Only then are you ready to contribute your own analysis.

Expanding Your Search Through Citation Chaining

Eventually, every database search reaches a point where new keyword combinations begin producing many of the same articles. When that happens, experienced researchers often stop thinking only about keywords and let the literature guide the next search. This approach is called **citation chaining**, sometimes referred to as **snowball searching** because one useful source can lead to many others.

Citation chaining follows the connections between scholarly publications in two directions: backward to earlier research and forward to newer research.

Backward Citation Chaining

Begin with a scholarly article that closely matches your research question. Then examine its reference list or bibliography. Each citation is a potential lead to an earlier study that informed the author's work. If the same articles or authors appear repeatedly in the

reference lists of multiple sources, they are often influential or foundational works in the field.

Forward Citation Chaining

Forward citation chaining begins with the same article but looks for newer publications that have cited it. Many library databases include a **Cited By** or **Times Cited** feature that lists later articles referencing the original work.

You can also use **Google Scholar** for forward citation chaining. Search for the article by its title, then locate it in the search results. Beneath the citation, select **Cited by** followed by a number (for example, *Cited by 127*). Google Scholar will display newer publications that cite the original article. You can further refine these results using keywords or the publication date filters to focus on the most relevant and recent research.

Using both backward and forward citation chaining helps you discover influential studies, identify major researchers in the field, and build a network of related scholarship that keyword searching alone might not uncover. One well-chosen article can quickly lead to dozens of additional sources.

Knowing When to Pivot

Every search provides feedback. Sometimes that feedback confirms that your search strategy is working. Other times, it signals that something needs to change. Learning to recognize these signals is an important part of the research process. Rather than repeating the same search over and over, successful researchers adjust their approach based on what they discover.

Too Few Sources

If careful searches produce only a handful of relevant sources, your topic may be too narrow, too recent, or too specialized for the assignment.

Try broadening your search by expanding the population, increasing the date range, or focusing on a broader aspect of the issue. You can also look for more general terms or related concepts that capture the same idea.

Too Many Sources

If your searches return thousands of relevant-looking results, your topic is probably too broad. Sorting through an overwhelming number of sources can be just as challenging as finding too few.

Narrow your search by adding a specific population, geographic location, setting, time period, or another limiting characteristic. A more focused research question usually produces a more manageable and relevant set of sources.

The Question Needs Revision

Sometimes the evidence suggests that your original research question is not the best one to ask. As you learn more about a topic, you may discover that an important assumption is unsupported or that another question is more interesting and better aligned with the available evidence.

Revising your research question is not starting over. The searching and reading you have already completed helped you understand the topic more deeply and guided you toward a stronger, more focused question.

Recognizing Saturation

As you continue searching, you may notice the same authors, landmark studies, and influential articles appearing repeatedly across different searches and reference lists. Researchers call this **saturation**. Reaching saturation does not mean you have found every source on your topic, but it does **suggest that you have identified the core scholarly conversation**.

Once you reach this point, continuing the same keyword searches often produces diminishing returns. Instead, shift your strategy. Use forward citation chaining to discover how newer researchers have built on, challenged, or refined these foundational studies. You can also spend more time reading the major works carefully, comparing perspectives, and identifying areas of agreement, disagreement, or unanswered questions.

Saturation is not a signal to stop researching. It is a sign that your search is becoming more focused and that you are ready to move beyond simply collecting sources toward understanding the conversation they represent.

Organizing Your Research

Good organization saves enormous amounts of time later in the research process.

As soon as you begin collecting sources, create a simple system for storing articles, recording notes, and tracking citations. Waiting until the end of a project often leads to missing citations, duplicate downloads, and unnecessary frustration.

There is no single "correct" way to organize your research. The best system is the one you will actually use consistently

The following approaches work well for most undergraduate researchers.

Google Docs and Cloud Storage

Many students find that a shared Google Drive folder and a Google Doc are all they need for a research project. Create one folder for your project and save PDFs, images, or other files there. Then create a Google Doc to record notes for each source, including:

- A complete citation
- A brief summary
- Important quotations (with page numbers)
- Keywords
- Questions or ideas for your paper

Because everything is stored in the cloud, your materials are available from any device and can be easily shared with instructors or classmates when appropriate.

Database Email and Project Folders

Most library databases allow you to email articles directly to yourself. Consider creating a dedicated email folder or label for each research project. As you find useful sources, email the citation, abstract, or full-text PDF to yourself and file the message in that folder.

This simple approach creates a searchable archive of your sources without requiring additional software. Just be sure to save important PDFs outside your email as well, since access to subscription databases may change after your course ends.

Citation Managers

Programs such as **Zotero** and **Mendeley** automatically save articles, organize collections, store notes, attach PDFs, and generate citations in multiple citation styles.

Although citation managers require a small investment of time to learn, they become increasingly valuable as research projects become more complex. If you expect to write multiple research papers during college, learning one of these tools is well worth the effort.

Regardless of which system you choose, begin organizing your research from the very first day. A few minutes spent organizing sources as you find them can save hours of searching, downloading, and recreating citations later.

Keeping Track of Your Searches

As your research develops, it can be difficult to remember which databases you searched, which keywords produced useful results, and which approaches led to dead ends. You do not need a formal search log, but it helps to keep a few notes as you work.

For example, you might record:

- Keywords that produced strong results
- Alternative terms or subject headings you discovered
- Databases that were most useful for your topic
- Searches that did not work, so you do not repeat them

These notes can be as simple as a page in your Google Doc or a note attached to your project folder. Spending a few minutes recording what you learned during each search will make it easier to refine your strategy the next time you sit down to research.

Ask a Professional

One of the most valuable research tools is another person.

Reference librarians, professors, and professionals in the field can recommend key resources, important researchers, useful keywords, and effective search strategies. A brief conversation can save hours of searching and help you avoid pursuing an unworkable research idea.

Experienced researchers routinely seek advice from colleagues. Developing the same habit will make your research faster, more focused, and more effective.

✓ Ask a Librarian

IVC Librarians are here to help!

We offer in person and online research assistance, including citation assistance. [Find our hours and more on the IVC Library website.](#)

Conclusion

Research rarely follows a perfectly planned path. Each search leads to new information, new questions, and better decisions about where to search next. By viewing research as an iterative process rather than a checklist, you become more flexible, more efficient, and ultimately more successful.

The strongest researchers are not those who find everything immediately. They are the ones who learn from each search, adapt their strategy, and steadily improve their understanding of the scholarly conversation.

Chapter Summary

Research is an iterative process in which searching, reading, evaluating, and revising occur repeatedly throughout a project. Rather than expecting the first search to produce perfect results, researchers use each search to improve the next one. Citation chaining expands a source pool by following references backward and citations forward, while saturation signals that the core literature has been identified. Researchers also recognize when to pivot by refining a research question or adjusting a search strategy. Finally, organizing sources, maintaining a search log, and seeking help from librarians make research more efficient and reduce frustration.

Key Concepts and Vocabulary

- **Iterative research.** A research process that continually adjusts questions, searches, and understanding as new evidence is discovered.
- **Citation chaining.** A search strategy that follows relationships between scholarly works through their citations.
- **Backward citation chaining.** Finding earlier sources by examining the reference list of a relevant publication.
- **Forward citation chaining.** Finding newer publications that cite a known source.
- **Saturation.** The point at which the same influential sources repeatedly appear across searches, indicating that the core literature has been identified.
- **Search log.** A record of databases searched, search terms used, filters applied, and results obtained to guide future searches.
- **Citation manager.** Software, such as Zotero or Mendeley, that stores sources, notes, and citations in an organized library.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 3.2A: Follow the Citation Trail

Choose one scholarly article related to your research question. Identify three useful sources from its reference list. Then, use the **Cited By** feature or Google Scholar to locate three newer articles that cite the original article.

Activity 3.2B: Evaluate Your Research Process

Review the searches you have completed for your current project and write an informal 250-word reflection addressing the following questions:

- What search strategy produced your best results?
- Did you need to revise your keywords or research question?
- Which sources or search techniques were least effective?
- What will you do differently during your next round of searching?

? Think About It: Discussion Questions

- Why do experienced researchers expect to revise their search strategy throughout a project, while many beginning researchers expect to find everything in their first search?
- Think about a time when changing your approach led to a better outcome in school, work, or everyday life. How is that experience similar to conducting iterative research?

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CHAPTER OVERVIEW

4: Analytical Reading and Evaluation

[4.1: Reading and Annotating Academic Texts](#)

[4.2: Evaluating Information](#)

[4.3: Logic, Reasoning, and Fallacies](#)

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4.1: Reading and Annotating Academic Texts

Unit 4 Analytical Reading & Evaluation | Chapter 4.1

Learning Objectives

- Identify the standard parts of a scholarly article (IMRaD).
- Apply a three-pass reading method to read research articles efficiently.
- Distinguish notes from annotations and use both purposefully.
- Write annotations that capture meaning, evidence, and your own questions.
- Convert annotations into a useful summary that you can draw on weeks later.

Introduction

Finding scholarly articles is only half the challenge. Once you've gathered relevant sources, you need to read them efficiently and understand what they're actually saying. Many students try to read journal articles from beginning to end, only to become overwhelmed by unfamiliar terminology, statistics, or dense academic writing.

The good news is that scholarly articles follow a predictable structure. Once you understand that structure, you can quickly identify the sections that matter most and spend your time where it counts. In this chapter, you'll learn how research articles are organized, how to use the three-pass reading method to read strategically, and how annotation can help you think critically, remember what you've read, and build a stronger foundation for your own research and writing.

The Anatomy of a Scholarly Article

Scholarly articles are not written to be read top to bottom. Once you know the architecture, you can read them faster, smarter, and with much better notes.

Most empirical research articles follow a structure called IMRaD — short for Introduction, Methods, Results, and Discussion. With an abstract at the top and references at the bottom, that's six standard sections. Knowing what each section is for tells you where to look when you need a specific piece of information.

The six standard sections of an empirical research article and the purpose of each section.

The Six Standard Sections of a Research Article	
Abstract	A 150–250 word summary of the entire paper. Read this first to decide whether the article is worth your time.
Introduction	Establishes the question, reviews relevant prior research, and states what this paper will contribute. The literature review lives here.
Methods	Explains how the study was conducted: who or what was studied, how data was gathered, and how it was analyzed. Skim for design; come back if you need to evaluate the methodology.
Results	Presents the findings, often with tables and figures. Look for the headline numbers and any unexpected patterns.
Discussion	Interprets the findings. What do they mean? What are the limitations? What's next? This is where the authors make their argument.
References	The bibliography. Don't skip this — it's the backward-citation goldmine you learned about in Chapter 3.2.

Not every scholarly article uses IMRaD strictly. Literature reviews, theoretical papers, humanities scholarship, and some qualitative work follow different structures. But IMRaD describes the majority of articles you'll encounter in the social sciences, sciences, and health fields.

The Three-Pass Reading Method

Reading a scholarly article from start to finish, top to bottom, is almost always the slowest way to do it. A more strategic approach is the three-pass method, originally developed by computer scientist S. Keshav (2007) and now widely adapted across disciplines. Each pass has a different purpose, and many articles only need one or two.

The three-pass reading method and the purpose of each stage of reading.

The Three Passes	
Pass 1 — Triage	5–10 minutes. Read the title, abstract, headings, and conclusion. Skim figures and tables. Decide: is this article relevant enough to keep reading?
Pass 2 — Mapping	30–60 minutes. Read the introduction and discussion carefully. Skim methods and results. Understand the argument and the evidence at a high level. Annotate as you go.
Pass 3 — Deep dive	1–3 hours. Read every section closely. Re-derive the methodology. Check the data. Question the conclusions. Reserved for the few articles you'll cite heavily.

Why Three Passes Save Time

If you treat every article as a Pass 3 read, you will quickly become overwhelmed. Most research projects require you to evaluate dozens of sources, but only a handful deserve a careful, in-depth reading. By beginning with Pass 1, you can quickly identify the articles most relevant to your research, reserve Pass 2 for the strongest candidates, and use Pass 3 only for the few sources you will rely on heavily. Effective researchers complete many Pass 1 readings, several Pass 2 readings, and very few Pass 3 readings.

Notes vs. Annotations

Both notes and annotations help you understand and remember what you read, but they serve different purposes. Notes are created separately from a source to organize information. Annotations are made directly on the source to support active reading and critical thinking. Most experienced researchers use both.

What Are Notes?

Notes are written records or summaries of information that you create in a notebook, document, or note-taking app. They may consist of phrases, complete sentences, outlines, diagrams, or lists organized in a way that makes sense to you. You might take notes during a lecture, while reading, or after a discussion.

Good notes are not transcripts. Instead, they identify the most important ideas, organize information with headings or categories, and emphasize key concepts that will be useful later.

What Are Annotations?

Annotations are comments, highlights, underlining, symbols, or questions added directly to a text, document, or image. Unlike notes, which are kept separately, annotations remain attached to the source itself. Their purpose is to help you interact with the material as you read rather than simply recording information.

Effective annotations serve several purposes:

1. **Explain** by interpreting important ideas or unfamiliar concepts.
2. **Clarify** by adding context or defining terms.
3. **Highlight** by marking evidence, key findings, or significant passages.
4. **Comment** by recording your own questions, observations, or critiques.

Common Forms of Annotation

- **Textual annotations:** Comments or notes written in the margins of a printed document or added as footnotes.
- **Digital annotations:** Highlights, sticky notes, comments, and bookmarks in PDF readers, Microsoft Word, Google Docs, or browser annotation tools.
- **Scholarly annotations:** Explanatory notes, citations, or references that scholars include to provide additional context or documentation.

Annotation Strategies That Work

Highlighting alone fails. Most students who only highlight read the article, mark up half of it in yellow, and remember nothing two weeks later. The trick is to make annotation an active practice — adding your own words, not just marking the author's.

Question-based annotation

As you read, write at least one question per page in the margins. "What does this term mean?" "Where is the evidence for this claim?" "How does this connect to what I read yesterday?" The questions force you to engage with the text, and they become a natural agenda for re-reading or for reaching out to a librarian.

Description vs. evaluation

Strong annotations distinguish what the source is saying (description) from what you think about it (evaluation). One useful trick is to use two different colors or symbols — one for paraphrase or summary, another for your own questions and reactions. When you come back to the source weeks later, you can tell at a glance which marks are about the article and which are about your thinking.

Why investing the time pays off

Writing annotations is not time lost; it is time invested. Spending the time to annotate carefully on the front end does three important things. First, it helps you prepare the ideas and content you will put into a research paper, and it ensures you recall which source to cite to support a claim. Second, it increases the likelihood that you will retain the information after the project is finished. Third, it saves you time later — your annotated notes will speed up future review, because you can revisit critical concepts quickly and efficiently.

From Annotations to a Useful Summary

Annotations help you engage with a source while you are reading. Summaries help you remember and use that source long after you finish reading it. After completing a careful read of an article, transform your annotations into two brief summaries that you can refer to throughout your research project.

Descriptive summary (about 200 words)

Summarize what the article says. Identify the research question, explain the methods, describe the major findings, and state the authors' conclusions. Focus on accurately representing the authors' ideas rather than offering your own opinions.

Evaluative summary (about 200 words)

Explain what you think about the article. Consider its strengths and limitations, how it compares with other sources you have read, and what questions or ideas it raises for your research. This is your opportunity to begin analyzing the source rather than simply describing it.

Together, these summaries become a valuable research tool. Instead of rereading every article each time you write, you can return to your summaries to recall the source's main ideas and your own insights. They will also become the foundation of your literature review in Chapter 5.2, where you will synthesize ideas across multiple sources.

Conclusion

Reading scholarly articles is a skill that develops with practice. The more you recognize an article's structure, read strategically, and annotate actively, the more efficient and confident you will become as a researcher. These habits will not only improve your current project but also prepare you for more advanced academic and professional reading.

Going Further: Reading Methods Critically

As your research skills develop, you will begin evaluating not only *what* researchers found but also *how* they reached their conclusions. Pass 3 reading asks you to examine the methodology itself. Was the sample appropriate? Were the measures valid? Do the data support the authors' conclusions? Answering these questions requires disciplinary knowledge that extends beyond this book, but you can begin developing those skills now.

For an honors extension, choose one article you have already completed through Pass 2. Read only the methods section carefully, noting each methodological decision the researchers made and one reasonable alternative they could have chosen. Then consider how that alternative might have influenced the study's findings.

Chapter Summary

Scholarly articles follow a recognizable structure that allows you to read strategically instead of reading every article from beginning to end. Understanding the IMRaD format helps you locate the information you need quickly, while the three-pass reading method ensures that you devote the most time to the sources that matter most. Notes and annotations serve different purposes, but both help you engage actively with a source and retain what you learn. By turning your annotations into descriptive and evaluative summaries, you create a lasting record of your research that will make writing your literature review and final paper much easier.

Key Concepts and Vocabulary

- **IMRaD.** The standard architecture of an empirical article: Introduction, Methods, Results, and Discussion, plus abstract and references.
- **Abstract.** The 150 to 250 word summary at the top of an article; read it first to decide whether to keep reading.
- **Three-pass method.** Triage, mapping, and deep dive: three readings with different purposes and time budgets (Keshav, 2007).
- **Notes.** Records kept separately from a source, organized so they make sense to you later.
- **Annotation.** A comment, question, highlight, or other mark added directly on the source itself.
- **Descriptive and evaluative summaries.** Paired 200-word paragraphs capturing what an article says and what you think about it.

Critical Thinking Activities

Try It: Reflection Activities

Activity 4.1A — Three-Pass Read of One Article

Pick one article from your source pool. Do a Pass 1 read (10 minutes max) and decide whether it deserves Pass 2. If yes, do a Pass 2 read with annotations. Submit a screenshot of your annotated PDF.

Activity 4.1C — IMRaD Map

Take a peer-reviewed article and map each of its sections to the IMRaD structure. If a section doesn't fit cleanly, note where the article diverges from the standard structure — and explain why that might be (e.g., a literature review rather than an empirical study).

Think About It: Discussion Questions

1. Most students treat every reading the same way, regardless of how important it is to their project. What stops people from triaging — and what would help break the habit?
2. Why does highlighting alone usually fail to help students remember what they read?

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4.2: Evaluating Information

Unit 4 Analytical Reading & Evaluation | Chapter 4.2

Learning Objectives

- Explain the importance of evaluating information and its role in information literacy.
- Apply established evaluation frameworks and strategies to assess the credibility of information sources.
- Distinguish between different forms of false or misleading information and recognize common techniques used to manipulate audiences.
- Evaluate news and AI-generated content using evidence-based verification practices.
- Use multiple sources to verify claims and make informed judgments about the reliability of information.

Introduction

Every day, you make decisions based on information. You choose which headlines to read, which social media posts to believe, which videos to watch, and which claims deserve your attention. Accurate reporting, honest mistakes, persuasive opinion, advertising, satire, and deliberate deception often appear side by side, making it increasingly difficult to know what deserves your trust. In a world where anyone can publish content in seconds and AI can generate convincing text, images, and video, evaluating information has become an essential life skill.

Information evaluation is broader than news. The same habits you use to judge a news article also help you evaluate scholarly research, websites, government reports, videos, and AI-generated content. This chapter introduces practical frameworks for evaluating information in any context before focusing on the additional challenges presented by today's news and social media environment.

Why Information Evaluation Matters

Information literacy is not simply finding information. It is deciding whether that information deserves your trust. A convincing presentation, professional-looking website, or viral social media post does not guarantee accuracy. Likewise, a source may contain valuable information while still reflecting bias, incomplete evidence, or persuasive intent.

Throughout this textbook you have developed skills in critical thinking, recognized different types of information sources, and considered how your own experiences shape your interpretation of evidence. Evaluation brings those skills together. Rather than accepting or rejecting information based on instinct or agreement, effective evaluators ask systematic questions about evidence, authority, context, and purpose.

The strategies in this chapter can be applied to nearly every kind of information you encounter, from scholarly articles and government reports to YouTube videos, podcasts, websites, and news stories.

The CRAAP Test

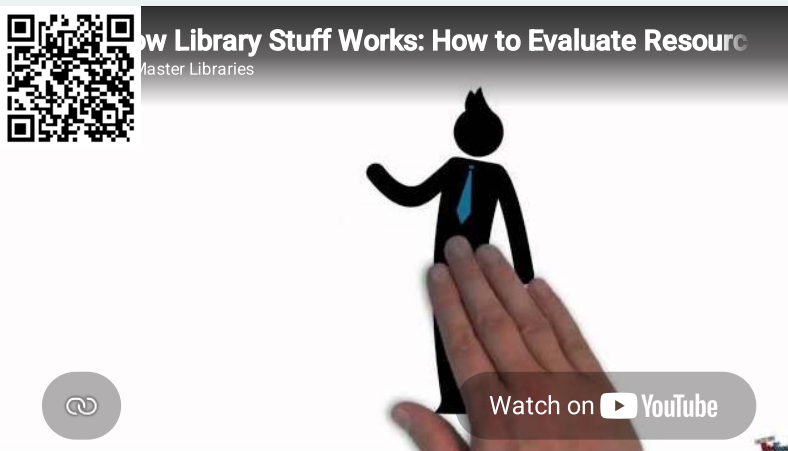
The CRAAP test is a five-criterion evaluation framework developed by librarian Sarah Blakeslee at California State University, Chico (Blakeslee, 2004). It works on almost any source: a website, a news article, a peer-reviewed paper, a YouTube video, a government report. The five letters stand for Currency, Relevance, Authority, Accuracy, and Purpose.

The CRAAP Test provides five criteria for evaluating the credibility, quality, and usefulness of information sources.

The CRAAP Test	
Currency	When was the information published or posted? Has it been revised or updated? Are the links functional?
Relevance	Who is the intended audience? Does the source meet the needs of your assignment? Is it written at an appropriate level — neither too elementary nor too specialized?
Authority	Who is the author, publisher, or sponsor? What are the author's credentials or organizational affiliations? What does the URL reveal about the source — .com, .edu, .gov, .org, .net?
Accuracy	Are there spelling, grammar, or typographical errors? Is the information supported by evidence? Are there citations or references? Has the work been reviewed by professionals?
Purpose	Is the source intended to inform, teach, sell, entertain, or persuade? Do the authors or sponsors make their intentions clear? Is the information fact, opinion, or propaganda?

Video Title

Watch this 2-minute video. Captions are available in the video player.



Make It a Habit

CRAAP works best as a habit rather than a checklist. The five criteria work together. A source may be current and relevant but lack authority. It may come from an authoritative organization but be written to persuade rather than inform. Strong evaluators consider all five criteria before deciding whether a source is appropriate for their purpose.

Looking Beyond the Source

The CRAAP Test evaluates information by examining the source itself. Sometimes, however, the fastest way to determine credibility is to leave the source and investigate it elsewhere.

Lateral Reading

Lateral reading is an evaluation strategy developed by digital literacy educator **Mike Caulfield** (2019, 2023). Instead of staying on a website and trying to determine whether it is trustworthy, lateral readers leave the page to investigate the source, author, or claim using independent sources.

Professional fact-checkers use this approach because it allows them to verify credibility more quickly and accurately than relying on the information presented by the website itself. Research by the Stanford History Education Group found that experienced fact-checkers consistently outperformed historians and college students by leaving unfamiliar websites to verify them elsewhere rather than reading them more carefully (Wineburg & McGrew, 2017).

The SIFT Method

Stop. Before accepting or sharing information, pause. Consider whether the claim is emotionally charged, surprising, or designed to provoke an immediate reaction.

Investigate the source. Open a new tab and learn about the author, organization, or website. Look for information about its mission, reputation, expertise, and funding rather than relying on what it says about itself.

Find better coverage. Search for the same claim in other credible sources. If no established organizations are reporting the story, or if only one source is making the claim, treat it with caution.

Trace claims to the original. Follow citations, links, images, statistics, or quotations back to their original source. Reading the original report, study, speech, or dataset helps you determine whether the information has been accurately represented or taken out of context.

SIFT is not a checklist to complete after you finish reading. It is a habit of verification that encourages you to leave a source, investigate it independently, and return only after you have established whether it deserves your trust.

Cross-Referencing and Triangulation

Triangulation means verifying important claims through multiple independent sources. Before accepting a surprising statistic, quotation, or news story, ask whether several credible organizations are reporting the same information.

For current events, a useful guideline is to seek confirmation from at least three independent, reputable news organizations. If you cannot find independent verification, treat the information as unverified rather than automatically accepting or rejecting it.

Find the Original

Claims often become distorted as they are repeated online. Whenever possible, locate the original report, research study, government document, speech, interview, image, or dataset rather than relying on summaries or screenshots. Reading the original source allows you to evaluate the evidence for yourself.

News Literacy

News literacy applies general information evaluation skills to journalism and current events. It helps you decide which reports deserve your trust while recognizing that news organizations operate under deadlines, incomplete information, and constant public scrutiny.

Professional journalism follows ethical standards designed to promote accuracy and accountability. The Society of Professional Journalists (2014) Code of Ethics encourages journalists to seek truth and report it, minimize harm, act independently, and remain accountable and transparent. Credible news organizations demonstrate these principles by publishing corrections, identifying sources whenever possible, distinguishing reporting from opinion, and explaining how information was obtained.

Understanding these standards gives you meaningful benchmarks for evaluating news coverage rather than relying solely on personal agreement or disagreement with a story.

Information Disorder

Information disorder refers to the spread of false, misleading, or harmful information. The concept recognizes that not all problematic information is the same. Some inaccuracies are accidental, some are intentionally deceptive, and some involve truthful information used in misleading or harmful ways. These three forms of information disorder are known as **misinformation**, **disinformation**, and **malinformation** (Wardle & Derakhshan, 2017).

Three Categories of Misleading Information

Misinformation

False or inaccurate information that is shared without the intent to deceive. The person sharing it may believe it is true or simply make an honest mistake. Misinformation is a problem of accuracy rather than intent.

Disinformation

False information that is deliberately created or shared to deceive, manipulate, or influence others. Disinformation is intentional and is often designed to spread quickly while resisting correction. What is commonly called "fake news" is often an example of disinformation.

Malinformation

Accurate or genuine information that is shared to cause harm, mislead through context, or violate privacy. Unlike misinformation or disinformation, the information itself may be true, but it is presented in a way that is intended to damage a person, organization, or community. Examples include publishing private emails without context, sharing personal information to harass someone, or recirculating an old photograph as though it depicts a current event.

The distinction matters because each category requires a different response. Misinformation can often be corrected by providing accurate information. Disinformation requires recognizing deliberate attempts to manipulate audiences and verifying claims through independent sources. Malinformation reminds us that even truthful information can be used unethically when it is stripped of context or shared to cause harm. As an information-literate reader, your goal is not only to determine whether information is true, but also to consider **how** and **why** it is being presented.

Common Traps in Online Information

Whether you are reading news, blogs, or social media, several recurring patterns deserve immediate attention.

- **Clickbait** uses sensational or emotionally charged headlines to attract attention and generate clicks. The headline often exaggerates or misrepresents the content, promising more than the article actually delivers.
- **Hoaxes** are fabricated stories or manipulated media created to deceive. They may include false news articles, edited photographs, deepfake videos, or fabricated documents designed to appear authentic.
- **Imposters** are websites or social media accounts that imitate legitimate organizations or news outlets. They often use similar names, logos, or web addresses to confuse readers. For example, **abcnews.com.co** was a fake website designed to resemble the legitimate **abcnews.go.com**. Always examine the full URL before trusting a source.
- **Disguised advertising** (sometimes called native advertising or sponsored content) presents promotional material in a format that resembles news or editorial content. Look for labels such as **Sponsored**, **Ad**, **Promoted**, or **Ad Choices**. Although these labels are often present, they may be subtle, making advertisements appear more like independent reporting.

Every piece of information serves some purpose. It may inform, persuade, entertain, generate advertising revenue, or provoke emotional reactions. Asking why content exists often reveals as much as examining what it says.

Evaluating AI-Generated Content

Artificial intelligence has dramatically increased both the quantity and quality of synthetic content available online. AI systems can produce convincing articles, images, audio recordings, and videos that appear authentic while containing fabricated information.

The evaluation strategies in this chapter remain effective because they focus on evidence rather than appearance. AI detection tools are unreliable and frequently produce false positives and false negatives. Instead of asking whether AI created the content, ask who published it, whether accountable authors stand behind it, and whether independent sources confirm its claims.

When evaluating AI-generated material, provenance matters more than technology. A claim supported by multiple credible sources deserves more confidence than an isolated image, quotation, or article that cannot be independently verified.

Eight Tips for Evaluating News

General evaluation skills apply to every source, but news presents unique challenges because information develops rapidly and often reaches audiences before all the facts are known. The International Federation of Library Associations (IFLA, 2017) developed an eight-step guide that summarizes many of the habits practiced by experienced news consumers.

1. **Consider the source.** Click away from the story to investigate the website, its mission, and its contact information.
2. **Read beyond the headline.** Headlines are often written to attract clicks and may exaggerate or oversimplify the story. Read the entire article before forming a conclusion.
3. **Check the author.** Search for the author's credentials and previous work. Are they a real person with relevant expertise?
4. **Examine the supporting sources.** Follow the cited links, references, or quotations. Do they actually support the claims being made?
5. **Check the date.** Old stories, images, or videos are often re-shared as though they are current events. Make sure the information is still relevant.
6. **Ask whether it might be satire.** If a story seems too outrageous to be true, consider whether it was intended as humor rather than factual reporting.
7. **Check your own biases.** Personal beliefs and assumptions can influence how you interpret information. Be willing to question your first reaction.
8. **Ask the experts.** Librarians, professional fact-checkers, and reputable fact-checking organizations can help you evaluate unfamiliar or questionable claims.

When uncertainty remains, ask a librarian. Reference librarians evaluate sources every day and can help you investigate unfamiliar publications, confusing claims, or difficult research questions.

News literacy is not a single technique but a collection of habits that become stronger with practice. The more often you question claims, verify evidence, and compare sources, the more confident and efficient you become at evaluating information. The following extension provides an opportunity to see how professional fact-checkers apply these same principles before you review the chapter's key ideas.

Introduction

Evaluating information is a skill that develops through practice, not memorization. The more often you question claims, compare sources, and verify evidence, the more confident and efficient you become at separating reliable information from misinformation and manipulation. The strategies in this chapter work best when they are used together, giving you multiple ways to assess credibility before accepting a claim as true. The following extension offers an opportunity to see how professional fact-checkers apply these same habits in their work before you review the chapter's key concepts.

Going Further

Professional fact-checkers at organizations such as **PolitiFact**, **Snopes**, and **FactCheck.org** publish both their conclusions and the evidence behind them. Their work provides useful models for evaluating claims rather than simply accepting a verdict.

For an honors extension, select a published fact-check and work backward through the investigation. What claim was examined? Which sources did the fact-checkers consult? How did they determine which evidence was credible? Finally, apply the same process to a claim related to your own research topic.

Chapter Summary

Evaluating information is one of the central skills of information literacy. Frameworks such as the CRAAP Test, lateral reading, triangulation, and finding original sources help you judge the credibility of information across many formats, from scholarly articles to websites and videos. News literacy applies these same evaluation skills to journalism and current events, where misinformation, disinformation, and rapidly changing information create additional challenges. Recognizing common online traps, understanding professional journalism standards, and evaluating AI-generated content all contribute to more informed decision making. The strongest evaluators rely on multiple strategies, verify important claims through independent evidence, and remain willing to question information before accepting it as true.

Key Concepts and Vocabulary

- **News literacy.** The ability to evaluate news and information sources well enough to decide what to trust.
- **Misinformation.** False information spread without intent to mislead; an honest error.
- **Disinformation.** Content created to purposefully mislead; what is colloquially called fake news.
- **Malinformation.** Genuine information deployed to cause harm, such as a real photo re-captioned to smear someone.
- **CRAAP test.** Five evaluation criteria: Currency, Relevance, Authority, Accuracy, and Purpose (Blakeslee, 2004).
- **Lateral reading.** Leaving a source to verify it through other sources, the way professional fact-checkers do.
- **Triangulation.** Confirming a story through at least three independent, credible outlets before trusting it.

Critical Thinking Activities

? Try It: Reflection Activities**Activity 4.2A — Apply CRAAP to Three Sources**

Choose three sources from your project that look credible at first glance: one from a news website, one from a database, and one from the open web. Apply CRAAP to each. Submit a short evaluation (about 150 words per source) noting which criteria each source passes strongly and which ones raise questions.

Activity 4.2B — Lateral-Read a Suspicious Story

Find a news story circulating on social media that you're uncertain about. Open a new tab. Investigate the publishing site, the author, and the central claim by searching for them outside the original story. Find at least two other credible sources covering the same story. Submit a 200-word write-up of what you learned.

? Think About It: Discussion Questions

1. Lateral reading goes against the natural instinct to keep reading the source you're already on. Why is the natural instinct so strong, and what helps you override it?
2. "Disinformation is built to evade correction." What does that mean for your responsibility as a reader, beyond just spotting individual fake stories?

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4.3: Logic, Reasoning, and Fallacies

Unit 4 Analytical Reading & Evaluation | Chapter 4.3

Learning Objectives

- Distinguish deductive from inductive reasoning.
- Distinguish among quantitative, qualitative, and anecdotal evidence.
- Identify common logical fallacies in arguments and media.
- Evaluate whether a conclusion logically follows from its evidence.

Introduction

Every research paper asks a question, gathers evidence, and reaches a conclusion. Logic is what connects those three steps. Without sound reasoning, even accurate facts can lead to weak or misleading conclusions. Learning to recognize how arguments are built helps you evaluate the claims you read and construct stronger arguments of your own.

In this chapter, you'll examine the two fundamental patterns of reasoning, distinguish among different types of evidence, recognize common logical fallacies, and learn how to draw conclusions that accurately reflect the available evidence. By the end of the chapter, you'll be better equipped to separate convincing arguments from merely persuasive ones and to ensure that your own conclusions follow where the evidence leads.

Deductive vs. Inductive Reasoning

Most arguments rely on one of two fundamental patterns of reasoning - deductive or inductive. Recognizing these patterns allows you to evaluate how conclusions are reached and whether the reasoning is logically sound.

Inductive Reasoning

Most empirical research relies on inductive reasoning. Researchers collect observations or data from a sample and use those findings to draw conclusions about a larger population. Because these conclusions extend beyond the individuals who were actually studied, they are always probable rather than certain. The strength of an inductive argument depends on how well the sample represents the broader population. This is why sample size, sampling method, and replication are essential for producing reliable conclusions.

Example: Suppose researchers survey 2,000 randomly selected college students from across the United States and find that 72% report using artificial intelligence tools for coursework. They may reasonably conclude that AI use is common among U.S. college students. However, if the study surveyed only 20 students from a single college, that same conclusion would be much less convincing because the sample would be too small and not representative of the larger population.

Deductive Reasoning

Most pure mathematics and formal logic rely on deductive reasoning, as does much of legal reasoning. Deductive reasoning begins with one or more general premises and applies them to a specific case. If the premises are true and the reasoning is logically valid, the conclusion must also be true. However, if even one premise is false, the conclusion is no longer guaranteed, regardless of how logical the argument appears.

Example: Consider the following argument. Because both premises are true and the reasoning is valid, the conclusion must be true. If the second premise were changed to "Dolphins are fish," the argument would no longer support a reliable conclusion because one of its premises is false.

- **Premise 1:** All mammals are warm-blooded.
- **Premise 2:** Dolphins are mammals.
- **Conclusion:** Therefore, dolphins are warm-blooded.

Definition: Premise

A **premise** is a statement that provides the basis for a conclusion. In a deductive argument, the conclusion follows logically from the premises. If the premises are true and the reasoning is valid, the conclusion must also be true.

Three Kinds of Evidence

Both deductive and inductive reasoning rely on evidence, but they use it in different ways. Deductive arguments use premises to establish conclusions that are logically certain when the premises are true. Inductive arguments use observations and evidence to support conclusions that are probable rather than certain. Because inductive reasoning extends beyond the evidence that was directly observed, the quality, quantity, and representativeness of that evidence are especially important.

Comparison of quantitative, qualitative, and anecdotal evidence, including their strengths, limitations, and appropriate uses in research.

Quantitative, Qualitative, and Anecdotal Evidence	
Quantitative	Numbers. Statistics, measurements, survey results, experimental data. Strengths: comparable, generalizable, lets you spot patterns across large populations. Limits: misses meaning, context, and the why behind the numbers.
Qualitative	Words and descriptions. Interviews, observations, case studies, ethnographies. Strengths: captures lived experience, surfaces meaning, reveals mechanisms. Limits: smaller samples, harder to generalize, more subject to interpretation.
Anecdotal	Personal stories or isolated examples. Strengths: vivid, relatable, useful for illustration. Limits: a sample of one (or a few) is not a basis for a general claim. Anecdotes feel persuasive but are usually weak evidence on their own.

Strong arguments often combine quantitative and qualitative evidence. Statistics tell you what is happening at scale; interviews and case studies tell you why. Anecdotes can introduce a topic or illustrate a point, but they should not be the foundation of a conclusion.

Common Logical Fallacies

A **logical fallacy** is a flaw in reasoning that makes an argument appear convincing even though its logic is unsound (Dowden, n.d.). Learning to recognize common fallacies helps you evaluate arguments more critically and identify weaknesses in reasoning. The following nine fallacies are among the most common and appear frequently in news reporting, social media, political discourse, advertising, and even academic writing.

Nine common logical fallacies, their definitions, and examples of how they appear in everyday and academic arguments.

Nine Common Fallacies

Nine Common Fallacies	
Ad Hominem	Attacking the person making the argument rather than the argument itself. "You can't trust her data, she's a graduate student."
Straw Man	Misrepresenting an opponent's argument to make it easier to defeat. "He says we should regulate emissions, so apparently he wants the economy to collapse."
False Dichotomy	Presenting only two options when more exist. "Either artificial intelligence will solve all our problems, or it will destroy society."
Slippery Slope	Claiming one event will inevitably lead to a chain of dire outcomes without showing the steps. "If we allow this, the next thing you know..."
Hasty Generalization	Drawing a broad conclusion from a small or unrepresentative sample. "My grandfather smoked his whole life and lived to 95 — so smoking can't be that bad."
Post Hoc	Assuming that because B followed A, A caused B. The classic correlation-isn't-causation error. "Crime fell after the new mayor took office, so the new mayor must be reducing crime."
Appeal to Authority	Citing an authority who isn't actually qualified on the question, or treating any expert as automatically correct. "My dentist says climate change isn't real."
Circular Reasoning	Smuggling the conclusion into the premises. "This source is trustworthy because it always tells the truth."
Red Herring	Introducing an irrelevant point to distract from the actual issue. "You ask about the budget shortfall — but what about all the good things we've accomplished?"

Spotting Fallacies As a Habit

Logical fallacies appear in many forms of communication, including campaign advertisements, social media posts, news commentary, and everyday conversations. Developing the habit of recognizing and naming these fallacies helps you evaluate arguments more critically and avoid being persuaded by flawed reasoning. It also strengthens your own writing by making you more aware of reasoning errors before they become part of your argument.

Drawing Logical Conclusions from Evidence

The conclusion of a research project is where logic gets tested most directly. Strong conclusions reflect what the existing research actually shows. They do not necessarily answer your original research question — and that is okay.

Three habits separate strong conclusions from weak ones.

Tie the conclusion to the evidence you reviewed

A research-paper conclusion is not the place to introduce new evidence or new claims. It synthesizes what your sources told you. If your conclusion makes a claim your sources don't support, that's a logic break — and a careful reader will catch it.

Acknowledge limitations honestly

Every body of research has gaps — populations not studied, time periods not covered, methodologies not used. Strong conclusions name those limits explicitly. This is not a sign of weak research; it is a sign of careful reasoning. Readers trust authors who acknowledge what they don't know.

Suggest further research where the evidence runs out

If your sources don't fully answer your question, say so — and propose what kinds of studies would help close the gap. This signals that you understand the conversation you're entering and points the next researcher toward useful work.

Challenge Assumptions

When the evidence doesn't agree with where you started

Sometimes the evidence leads to a different conclusion than the one you expected when you first developed your research question. Effective researchers do not reshape evidence to support their original assumptions. Instead, they allow the evidence to guide their conclusions, even when it challenges their initial perspective. Revising your position in response to credible evidence demonstrates intellectual humility (Chapter 1.2) and thoughtful reflection (Chapter 1.3). Changing your mind in light of better evidence is not a weakness; it is a hallmark of sound research and critical thinking.

Conclusion

Reasoning is what transforms information into knowledge. Research is not simply a process of collecting sources or quoting experts; it is the process of using evidence to reach conclusions that are logical, well supported, and appropriately qualified. Strong researchers understand that evidence does not speak for itself. They evaluate its quality, recognize the limits of what it can demonstrate, and avoid reasoning errors that can undermine an otherwise persuasive argument.

The habits explored in this chapter, including recognizing deductive and inductive reasoning, evaluating different forms of evidence, identifying logical fallacies, and acknowledging the limits of research, form the foundation of sound academic writing. As you continue through the research process, these habits will help you build arguments that are not only persuasive, but also intellectually honest and worthy of your readers' trust.

Going Further: Fallacies in Scholarly Writing

Fallacies are easy to spot in campaign ads and much harder to spot in polished academic prose, where they hide inside hedged language and impressive citations. Overgeneralization from small samples and post hoc causal claims are the most common. For an honors extension, take one peer-reviewed article from your source pool and audit its reasoning: identify its strongest inference and the one place where its conclusion outruns its evidence.

Chapter Summary

Logic and reasoning are the structural integrity of an argument. Deductive reasoning moves from general principles to certain conclusions; inductive reasoning moves from specific observations to probable ones. Evidence comes in three forms — quantitative, qualitative, and anecdotal — each with its own strengths and limits. Nine common fallacies (ad hominem, straw man, false dichotomy, slippery slope, hasty generalization, post hoc, appeal to authority, circular reasoning, and red herring) capture the most frequent ways arguments break. Strong research conclusions tie themselves tightly to the evidence reviewed, acknowledge limitations honestly, and follow the evidence even when it changes the writer's mind.

Key Concepts and Vocabulary

- **Deductive reasoning.** Top-down reasoning from a general principle to a specific case; certain if the premises are true.
- **Inductive reasoning.** Bottom-up reasoning from specific observations toward a probable general conclusion.
- **Quantitative, qualitative, and anecdotal evidence.** Numbers at scale; words and lived experience; and isolated personal stories.
- **Logical fallacy.** A pattern of broken reasoning that looks like it works but does not.
- **Post hoc fallacy.** Assuming that because B followed A, A caused B; the classic correlation-causation error.
- **Limitations.** The honest naming of what a body of research does not cover.
- **Premise.** A statement that provides the basis for a conclusion.

Critical Thinking Activities

Try It: Reflection Activities

Activity 4.3A — Fallacy Hunt

Spend 30 minutes in current media — news commentary, opinion pieces, political ads, social media. Find five examples of fallacies from this chapter. For each, write one sentence describing the fallacy and one sentence explaining what a sound version of the argument would look like.

Activity 4.3B — Map Your Evidence

List the sources in your current source pool. Categorize each as primarily quantitative, qualitative, or anecdotal. Are you over-relying on one type? What kinds of evidence might be missing — and where could you find them?

Think About It: Discussion Questions

1. Why do anecdotes feel so persuasive even when they are weak evidence? What does that tell you about how human beings reason?
2. "Strong conclusions don't necessarily answer the original research question." Why not? What's the alternative?

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CHAPTER OVERVIEW

5: Synthesis and Scholarly Conversation

[5.1: Synthesizing Sources and Joining the Conversation](#)

[5.2: Writing the Literature Review](#)

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5.1: Synthesizing Sources and Joining the Conversation

Unit 5 Synthesis & Scholarly Conversation | Chapter 5.1

Learning Objectives

- Distinguish summary from synthesis.
- Identify themes, agreements, and tensions across multiple sources.
- Build a synthesis matrix to map your sources.
- Write paragraphs organized by theme rather than by source.
- Use signal phrases to integrate multiple voices into your own argument.

Introduction

By the time you've gathered and evaluated your sources, you've already done a significant amount of work. The next challenge is figuring out what all of those sources say together. This is where many students struggle. Instead of comparing ideas across sources, they write one summary after another, producing a paper that reads like a list of article reviews rather than an analysis of the research.

Synthesis is the skill that transforms a collection of sources into a coherent discussion. Rather than asking, *What does this article say?*, synthesis asks, *What does the body of research tell us?* In this chapter, you'll learn how to identify themes across your sources, organize your notes with a synthesis matrix, and use that organization to write literature review paragraphs that compare, contrast, and connect ideas.

Summary vs. Synthesis

Although summary and synthesis are closely related, they serve different purposes in academic research and writing.

Summary

A **summary** explains what **one source** says in your own words. It condenses the author's main ideas while remaining faithful to the original meaning. A good summary accurately represents the source without adding your own analysis or interpretation. Because it focuses on a single source, it answers the question, "**What does this author say?**"

Synthesis

A **synthesis** goes a step further. Rather than describing one source at a time, it brings **multiple sources** into conversation with one another. A synthesis identifies patterns, agreements, disagreements, trends, and gaps across the research. Instead of simply reporting what each author says, it helps readers understand what the body of evidence suggests as a whole. In other words, synthesis answers the question, "**What do these sources collectively tell us that no single source could?**"

Think of it this way: **summary is about understanding individual voices; synthesis is about understanding the conversation among those voices.** Strong research papers include both. You summarize sources accurately, then synthesize them to build your own evidence-based argument.

Identifying Literature Themes

Themes

- A theme is simply a big idea discussed by more than one source.
- Themes emerge from the literature, not from your assumptions.
- Choose short, descriptive names for your themes because they often become the section headings in your literature review.

Synthesis begins by identifying the ideas that appear repeatedly across your sources. These recurring ideas are called **themes**, and they become the organizational framework for your literature review. **Themes emerge from the literature, not from your assumptions.**

As you read and annotate your sources, look for recurring topics such as causes, effects, populations, research methods, proposed solutions, or unresolved questions. Three to five well-developed themes are usually enough for an undergraduate research paper.

A theme is simply a big idea discussed by more than one source. For example, if you're researching social media and misinformation, several authors may discuss how misinformation spreads rapidly on platforms like TikTok or X. Others may focus on declining public trust in news or on strategies for fact-checking. Each of these recurring topics represents a potential theme.

Naming Themes

Choose short, descriptive names for your themes because they often become the section headings in your literature review.

Example 5.1.1

Literature Review Themes

- Spread of Misinformation

- Public Trust and Media
- Fact-Checking on Social Platforms

As you read, the following questions will help you identify the themes that organize your research.

- What topics appear repeatedly?
- Which problems or questions do multiple authors discuss?
- Where do researchers agree?
- Where do they disagree?
- What questions remain unanswered?

Using a Synthesis Matrix

One of the biggest challenges in research is moving beyond writing one summary after another. A **synthesis matrix** helps you organize your notes so you can compare sources instead of treating each one separately. It is one of the most useful planning tools for writing a literature review, research paper, or annotated bibliography.

A synthesis matrix is simply a table. **Each row represents one source, and each column represents a theme, question, or idea that appears across multiple sources.** As you read, you fill in each cell with a brief note about what that source says about the corresponding theme.

For example, imagine you are researching the effects of social media on college students.

Example of a synthesis matrix for organizing research findings across multiple sources.

Source	Causes of the Problem	Effects on Students	Suggested Solutions
Smith (2021)	Platform algorithms encourage constant engagement	Increased anxiety	Limit daily social media use
Lopez (2022)	Peer pressure and social comparison	Sleep disruption	Create school-based digital wellness programs
Patel (2023)	Addictive content design	Anxiety and sleep disruption	Increase regulation of social media platforms

The real value of the matrix comes from how you read it.

- **Read across a row** to remind yourself what **one source** says. This is essentially a summary.
- **Read down a column** to compare what **all of your sources** say about a single theme. This is where synthesis begins.

✓ Example 5.1.2

A Literature Review Sentence

Look only at the **Effects on Students** column, you can quickly see that multiple researchers report anxiety, sleep disruption, or both. That comparison allows you to write craft a sentence.

Across the literature, researchers consistently found that heavy social media use is associated with negative mental health outcomes, particularly anxiety and sleep disruption, although they differ in which effects they emphasize.

Notice that this sentence does not belong to any one source. Instead, it combines evidence from several sources into a single idea. That is synthesis.

A synthesis matrix also helps you identify **gaps and disagreements**. You may notice that every source discusses the effects of a problem, but only one offers practical solutions. Or two researchers may agree that social media affects mental health but disagree about what causes the problem. Those patterns are often the most interesting findings in your research because they point to questions that deserve further investigation.

As your project grows, your matrix becomes a roadmap for your paper. Instead of organizing your writing around individual authors, you can organize it around the major themes that emerge from the research.

From Matrix to Outline

Once you've completed your synthesis matrix, you've done much of the hard work of organizing your research. The next step is to turn that organized information into an outline for your paper. Instead of planning your paper around individual sources, you'll organize it around the **themes** that emerged from your reading.

Think of your synthesis matrix as a bridge between **reading** and **writing**. The matrix helps you discover the major conversations in the literature. Your outline transforms those conversations into the structure of your paper.

Suppose your matrix identified three recurring themes:

- Causes of the problem
- Effects on students
- Proposed solutions

Those themes naturally become the major sections of your literature review.

✓ Example 5.1.3

Literature Review Outline

I. Causes of the Problem

- Compare what multiple researchers identify as the major causes.

II. Effects on Students

- Synthesize the evidence describing the consequences.

III. Proposed Solutions

- Compare recommendations and identify areas of agreement and disagreement.

Notice that the paper is organized by **ideas**, not by **authors**. Instead of writing one paragraph about Smith, one about Lopez, and one about Patel, each section weaves together evidence from several sources to explain a single theme.

Your synthesis matrix becomes a roadmap. Each column suggests a section of your outline, and the notes within that column become the evidence for your synthesis paragraphs.

They Say, I Say: Entering the Conversation

Synthesis only works if you treat your sources as voices in a conversation rather than as decorations for your own argument. Gerald Graff and Cathy Birkenstein describe this approach as the "**They Say / I Say**" framework. Strong academic writing first explains what others have argued ("They say...") before presenting the writer's own contribution ("I say...").

As you compare your sources, pay attention to three common patterns:

- **Agreement:** Multiple researchers reach similar conclusions. These areas provide the strongest foundation for your discussion.
- **Disagreement:** Researchers interpret the evidence differently or reach conflicting conclusions. These debates often become the most interesting parts of a literature review.
- **Gaps:** Important questions remain unanswered or under-explored. Identifying gaps demonstrates that you understand where further research is needed.

Rather than thinking of sources as separate articles, imagine them participating in an ongoing scholarly conversation. Your role is to explain that conversation clearly and contribute your own evidence-based interpretation.

Writing Theme-Based Paragraphs

The most common synthesis mistake is organizing a paper **source by source** rather than **theme by theme**.

A **source-based paper** sounds like this: "*Smith says... Lopez says... Patel says...*" Although accurate, this approach produces a series of summaries instead of a synthesis.

A **theme-based paper** organizes paragraphs around ideas rather than authors. Each paragraph begins by introducing a theme. It then integrates evidence from several sources, compares their findings, highlights similarities or differences, and ends by drawing a conclusion about what the research collectively suggests.

Readers should come away understanding the **state of the research**, not simply remembering which author said what.

Paraphrase over quotation

Most of your literature review should rely on **paraphrasing** rather than direct quotation.

Paraphrasing requires you to explain a source in your own words while preserving its original meaning. Because paraphrasing demands genuine understanding, it demonstrates your comprehension of the research.

✋ Limit Quotes

Reserve direct quotations for situations where the author's exact wording matters, such as a formal definition, a particularly influential statement, or language that cannot be paraphrased without losing its meaning

Remember that **every paraphrase still requires a citation**. Writing in your own words does not eliminate the need to acknowledge the original source.

Signal Phrases and Attribution

Signal phrases help readers understand where ideas come from while clarifying each source's role in the conversation. Different verbs communicate different relationships between authors and ideas.

Examples of signal-phrase verbs organized by the purpose they serve in academic writing.

Signal-Phrase Verbs by Stance

Neutral reporting

states, describes, explains, reports, observes, notes

Signal-Phrase Verbs by Stance

Argument or claim	argues, contends, maintains, asserts, posits, proposes
Investigation or finding	found, investigated, demonstrated, showed, documented
Push or critique	challenges, questions, complicates, critiques, takes issue with
Agreement or extension	extends, builds on, agrees, confirms, corroborates

Using a variety of signal verbs creates a more accurate picture of the scholarly conversation and helps readers distinguish between reporting evidence, making claims, and evaluating ideas.

Conclusion

Synthesis is the point where research becomes more than a collection of sources. By identifying recurring themes, organizing your notes in a synthesis matrix, and writing theme-based paragraphs, you help readers see the larger patterns in the literature rather than a series of disconnected summaries. This approach not only makes your writing more organized, but also demonstrates that you understand how individual studies relate to one another.

As you continue developing your research paper, remember that your goal is not simply to report what each author said. Your job is to explain what the research, taken as a whole, reveals about your topic. That ability to connect ideas across sources is one of the defining characteristics of strong academic writing.

Going Further: Meta-Analysis and Systematic Synthesis

The synthesis matrix introduces the same thinking researchers use in systematic reviews and meta-analyses. These forms of scholarship examine dozens or even hundreds of studies using transparent, reproducible methods to answer a focused research question.

As an extension activity, locate a systematic review related to your topic. Compare its methods section with your own synthesis process. What similarities do you notice? What additional procedures did the researchers use to make their review more comprehensive and reliable?

Chapter Summary

Synthesis is the process of moving from describing individual sources to explaining what the research collectively reveals. The process begins by identifying recurring themes across the literature, organizing those themes in a synthesis matrix, and using the matrix to build a theme-based outline. Strong literature reviews compare ideas rather than authors, treating sources as participants in a scholarly conversation. By relying primarily on paraphrase, using clear signal phrases, and organizing paragraphs around themes, you can transform a collection of summaries into a coherent, evidence-based discussion of the research.

Key Concepts and Vocabulary

- **Synthesis.** Pulling multiple sources together to say something no single source said.
- **Synthesis matrix.** A table with sources as rows and themes as columns; reading down a column is the raw material of synthesis.
- **Theme.** A recurring topic that emerges across multiple sources.
- **They say / I say.** Graff and Birkenstein's (2018) framework for naming what others have said and positioning your response.
- **Signal phrase.** A short clause such as "Smith argues" that attributes an idea and conveys your stance toward it.
- **Paraphrase.** Restating a source in your own words, with citation; the workhorse of research writing.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 5.1A — Build a Synthesis Matrix

Take five of the strongest sources in your source pool. Identify three themes that appear across at least three of them. Build a synthesis matrix (sources as rows, themes as columns).

Activity 5.1B — Map a Tension

Find a column of your synthesis matrix where two or more sources disagree. In 250 words, describe the disagreement: what does each source claim, what evidence does each rest on, and what would need to be true for the disagreement to be resolved?

? Think About It: Discussion Questions

1. Why do you think source-by-source organization is so common in early student writing? What does it feel safer about?
2. Why might over-quoting actually hurt the credibility of a research paper, even when the quotations are accurate?

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5.2: Writing the Literature Review

Unit 5 Synthesis & Scholarly Conversation | Chapter 5.2

Learning Objectives

- Describe the purpose and structure of a literature review.
- Choose an organizational structure that fits your topic — thematic, chronological, methodological, or theoretical.
- Use headings, transitions, and visual cues to guide a reader through your synthesis.
- Identify gaps in the existing research and articulate how your work fits into the conversation.

Introduction

By the time you begin writing a literature review, you have already completed much of the groundwork. You have explored your topic, developed a focused research question, located credible sources, evaluated their quality, and synthesized their ideas. Now it is time to turn that preparation into a coherent piece of academic writing.

A literature review is more than a collection of summaries. It organizes existing scholarship into meaningful patterns, identifies areas of agreement and disagreement, and explains how your research fits into the broader scholarly conversation. Instead of presenting one source after another, you guide readers through what the field knows, what it still debates, and where your research can contribute.

In this chapter, you will learn how to organize, write, and revise a literature review that establishes a strong foundation for the rest of your research paper.

Understanding the Purpose of a Literature Review

A literature review is not a book report or a series of source summaries. It is a synthesis of the scholarly conversation surrounding your research question. Rather than discussing one source at a time, you identify patterns across the literature, highlight areas of agreement and disagreement, and explain what questions remain unanswered. Your goal is not to include every source you read, but to organize the most relevant research into a clear, coherent argument about what the literature reveals.

A strong literature review establishes the foundation for the rest of your research paper. Before readers encounter your methods, analysis, or conclusions, they should understand what the field already knows, where scholars disagree, and what gap your research is designed to address. Without that context, your argument lacks the scholarly grounding that makes it persuasive.

Think of yourself as a guide rather than a reporter. Your job is to help readers make sense of the research by showing how the pieces fit together and why that conversation leads naturally to your own research question.

Choosing an Organizational Structure

A well-organized literature review helps readers see patterns in the research rather than a series of disconnected studies. The way you organize your review should reflect the story the literature tells. While there is no single correct approach, four organizational structures are commonly used. The best choice depends on your research question, the kinds of sources you have collected, and the relationships you want to emphasize.

Organizing a Literature Review. Thematic, chronological, methodological, and theoretical structures each emphasize different relationships within the scholarly conversation.

Four Common Organizational Structures	
Thematic	Most common. Organize around the major themes that emerge across your sources (Chapter 5.1). Each section addresses one theme and weaves multiple sources through it. Best when sources cluster around shared topics.
Chronological	Organize by time period. Best when the literature shows how thinking on a topic has evolved — early work, later work, current work — or when historical context matters to the argument.
Methodological	Organize by research method. Best when different methods (qualitative vs. quantitative, surveys vs. interviews) produce meaningfully different findings on the same question.
Theoretical	Organize around the theoretical frameworks researchers have used. Best when the field is shaped by competing theories more than by competing findings.

From Planning to Drafting the Literature Review

Once you have chosen an organizational structure, you can begin drafting your literature review. Although every project is different, effective literature reviews share a common architecture. As you write, your goal is to move readers through the scholarly conversation in a logical sequence, ending with a clear explanation of how your own research contributes to that conversation.

The Structure of a Literature Review

Most undergraduate literature reviews contain four major parts.

1. Introduction

Begin by introducing your research question and explaining why it matters. Provide enough background to orient readers, then briefly preview how the literature review is organized.

2. Body

Organize the body according to the structure you selected: thematic, chronological, methodological, or theoretical. Each section should synthesize multiple sources rather than discussing one source at a time. End each section with a brief takeaway that reinforces the main idea before transitioning to the next.

3. Gap Statement

After surveying the literature, explain what remains unknown. The gap may involve an unanswered question, an understudied population, conflicting findings, or a limitation in existing methods. Connect this gap directly to your own research question.

4. Conclusion

Conclude by summarizing what the literature reveals and explaining why your research is a logical next step. Readers should leave the literature review understanding both the current state of knowledge and the contribution your study will make.

Finding a Meaningful Research Gap

The gap statement is the turning point of a literature review. It explains why another study is needed by showing what the existing scholarship has not yet answered. Most gaps are not explicitly labeled; they emerge only after comparing multiple sources and looking for patterns.

Several places can help you identify a meaningful gap.

- **Study limitations and future research.** Researchers often acknowledge what their own studies could not address. When several authors recommend similar directions for future research, you have identified a promising opportunity.
- **Your synthesis matrix (Chapter 5.1).** Themes that appear frequently but receive little direct investigation often reveal areas that deserve additional study.
- **Unresolved disagreements.** When researchers reach different conclusions about the same question, additional evidence or a different research design may be needed to resolve the debate.
- **Missing populations, settings, or methods.** Findings based on one population, geographic region, or research method may not apply elsewhere. Extending existing research can be just as valuable as answering an entirely new question.

Not every gap is worth pursuing. A useful gap should matter to the field. It should have the potential to deepen understanding, improve practice, resolve a disagreement, or expand knowledge to new contexts. When writing your gap statement, explain both **what is missing** and **why filling that gap matters**.

Guiding Your Reader

Strong literature reviews are easy to follow because they guide readers from one idea to the next. Headings and transitions provide the signposts that make the organization visible.

Use a simple heading hierarchy throughout your review. Major headings should correspond to your primary themes, time periods, methods, or theoretical perspectives. Subheadings can be added when a section naturally divides into smaller ideas, but avoid creating unnecessary levels of organization.

Transitions are equally important. At the end of each section, briefly summarize the main point before introducing the next one. For example:

While most studies focus on individual social media habits, a smaller body of research examines institutional responses. The next section explores these approaches.

These brief connections help readers see the literature review as one continuous argument rather than a series of separate summaries.

Revising for Synthesis

Nearly every first draft of a literature review reads like a collection of individual summaries. Revision is the stage where synthesis becomes visible. A simple three-pass strategy can help strengthen your draft.

Coherence pass. Confirm that each paragraph develops a single idea and that sources are grouped by theme rather than presented one at a time.

Voice pass. Make sure your own analytical voice guides the discussion. Your topic sentences, transitions, and concluding statements should explain how the sources relate instead of allowing quotations and paraphrases to dominate the page.

Citation pass. Verify that every quotation and paraphrase is properly cited, quotations are used sparingly, and your citations consistently follow the required documentation style.

A Worked Example

The following paragraph illustrates how synthesis looks in a thematic literature review.

Researchers broadly agree that heavy nighttime social media use is associated with poorer sleep in adolescents, but they disagree about the mechanism. Smith (2021) attributes the effect to platform algorithms that reward late-night engagement, while Lopez (2022) points to peer pressure that keeps students responsive long after midnight. Patel (2023) complicates both accounts, reporting that content design, not timing alone, predicts disrupted sleep. Taken together, these studies suggest that limiting hours of use may matter less than changing what platforms show; no study to date has tested the two interventions against each other.

Notice how the paragraph works. The opening sentence introduces a **theme** rather than a source. Each study is incorporated using a purposeful signal phrase that explains its contribution to the conversation. The final sentence moves beyond summary by drawing a conclusion across the sources and identifying a meaningful research gap. This concluding synthesis is what distinguishes a literature review from a sequence of individual source summaries.

Conclusion

Writing a literature review is one of the most challenging parts of the research process because it asks you to do more than summarize sources. It asks you to recognize patterns, evaluate evidence, and explain how the scholarly conversation leads to your own research question. These skills take practice, but they also mark the transition from gathering information to contributing to knowledge. As your research projects become more advanced, the literature reviews you write will become longer and more sophisticated, but the principles of synthesis, organization, and critical analysis will remain the same.

Going Further: Extending the Review

If you are working on an extended or honors version of a research project, your literature review may be longer, more comprehensive, and may include additional moves: a methods section explaining how you searched, a positionality statement (Chapter 1.3) integrated into the review, or a critical-reflection appendix that reflects on the research process itself. These extensions are common in upper-division and graduate work, and the chapters in this book have prepared you for them.

Chapter Summary

A literature review maps a scholarly conversation and places your research question inside it. Choose an organizational structure (thematic, chronological, methodological, or theoretical) that fits your sources. Write a clear introduction, theme- or method-organized body sections, a gap statement, and a conclusion that points to your contribution. Use headings and transitions to guide the reader. Revise for coherence, voice, and citation accuracy. The literature review is not the easiest part of a research project — but it is the part that makes everything else credible.

Key Concepts and Vocabulary

- **Literature review.** A synthesis that maps a scholarly conversation and places your research question inside it.
- **Organizational structures.** Thematic, chronological, methodological, and theoretical: the four common ways to organize a review.
- **Gap statement.** The passage naming what the existing literature has not answered and connecting it to your question.
- **Transition.** One or two sentences at a section boundary that turn separate sections into a single argument.
- **Coherence, voice, and citation passes.** Three revision passes that turn stapled-together summaries into synthesis.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 5.2A — Outline Your Review

Build a complete outline of your literature review: introduction, three or four body sections (with subsections if needed), gap statement, and conclusion. For each section, list the sources you'll draw on and the takeaway you want the reader to leave with.

Activity 5.2C — Write a Gap Statement

In 250 words, write a gap statement: what is missing from the existing literature on your topic, and how does your research question (Chapter 2.3) speak to that gap?

? Think About It: Discussion Questions

1. Why is thematic organization the most common structure for undergraduate literature reviews? What does it ask of you that the other structures don't?
2. "A literature review is not a book report." What's the practical difference, and where does that difference show up in the writing?

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Association of College and Research Libraries. (2015, February 9). *Framework for information literacy for higher education*. American Library Association. <https://www.ala.org/acrl/standards/ilframework>

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CHAPTER OVERVIEW

6: Ethical Use and Communication of Research

[6.1: Ethical Use of Information and APA 7 Citation](#)

[6.2: Information Ethics and Intellectual Property](#)

[6.3: Communicating Research- Presentations, Posters and Research Abstracts](#)

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6.1: Ethical Use of Information and APA 7 Citation

Unit 6 Ethical Use & Communication of Research | Chapter 6.1

Learning Objectives

- Explain why academic integrity is essential to scholarship and research.
- Distinguish among direct, mosaic, accidental, and self-plagiarism.
- Choose appropriately between quotation, paraphrase, and summary when incorporating sources.
- Determine when information is considered common knowledge and when a citation is required.
- Apply the basic principles of APA 7 in-text citations and reference lists.
- Evaluate the strengths and limitations of citation management tools and AI-assisted citation generation.

Introduction

Research is built on trust. Readers trust that your evidence is accurate, your conclusions are supported, and the ideas you borrow from others are properly acknowledged. Academic integrity is the foundation of that trust. Every article you read, every statistic you cite, and every quotation you include represents someone else's time, expertise, and intellectual effort. Ethical research means recognizing those contributions while demonstrating your own thinking.

This chapter explains how scholars use sources responsibly. You will learn how plagiarism occurs, how to avoid it, when citations are required, and how to document sources using APA style. These practices are more than formatting rules. They are habits of honesty that strengthen both your writing and your credibility.

Citations and Academic Integrity

Citation is more than an academic requirement. It is how scholars acknowledge the work of others, support their own claims with credible evidence, and allow readers to verify where information came from. Every source you use represents someone else's time, expertise, creativity, and intellectual effort. Giving proper credit is both an ethical responsibility and a hallmark of trustworthy scholarship.

When you cite a source, you accomplish several important goals:

- **You give credit** to the original creator.
- **You allow readers to verify** your evidence.
- **You demonstrate** that your conclusions are supported by credible research.
- **You contribute** honestly to the ongoing scholarly conversation.

Citation is one of the defining characteristics of academic writing. Rather than presenting every idea as your own, scholars openly acknowledge the work that informed their thinking. This transparency builds trust, allows knowledge to be evaluated and expanded, and advances research as a collaborative enterprise.

Plagiarism and How to Avoid It

Plagiarism occurs whenever you present another person's words, ideas, data, images, or creative work as your own without appropriate acknowledgment. While many people think plagiarism always involves intentional cheating, most cases among undergraduate students are accidental. Poor note-taking, rushed writing, and incomplete citations often lead students to misuse sources without realizing it.

Fortunately, plagiarism is almost always preventable through careful research habits and consistent citation practices.

Forms of Plagiarism

Plagiarism can take several forms, and not all involve intentionally copying another person's work. Understanding the differences helps you recognize and avoid academic misconduct.

Direct plagiarism occurs when you copy another author's words exactly as they appear without quotation marks and without a citation. This is the most obvious and serious form of plagiarism because it presents someone else's writing as your own.

Mosaic plagiarism, sometimes called *patch writing*, occurs when you combine phrases, sentence structures, or ideas from one or more sources without proper attribution. Even if individual words are changed, writing that closely follows the original source without citation is still plagiarism.

Accidental plagiarism occurs when writers unintentionally fail to credit their sources. Incomplete notes, forgotten citations, or paraphrases that remain too close to the original wording are common causes. Although there may be no intent to deceive, accidental plagiarism is still a violation of academic integrity.

Self-plagiarism occurs when you submit work that you previously completed for another course without permission or disclosure. Although the work is your own, presenting previously submitted work as new violates the expectation that each assignment represents original effort.

Most plagiarism can be prevented by maintaining careful research notes and writing from your own understanding rather than copying directly from sources.

Three Ways to Use a Source

Academic writers generally rely on three techniques when incorporating information from sources: **quotation, paraphrase, and summary**. Each serves a different purpose, and knowing when to use each one helps you write more effectively and ethically.

Quotation uses a source's exact words enclosed in quotation marks, followed by a citation. Quotations are most effective when the original wording is especially important, such as a precise definition, a memorable statement, a controversial claim, or language whose wording would lose meaning if rewritten. Because quotations should be used selectively, they work best when they add something that your own wording cannot.

Paraphrase restates a specific idea or passage from a source entirely in your own words and sentence structure while preserving the original meaning. A citation is still required because the idea belongs to the original author. Paraphrasing is the technique you will use most often in academic writing because it demonstrates that you understand the material and can explain it clearly to your readers.

Summary condenses the main ideas of a longer text into a much shorter explanation written in your own words. Like paraphrasing, a summary must also be cited. Summaries are useful when readers need a broad understanding of a source's overall argument, findings, or purpose before you focus on specific evidence or details. They provide context without overwhelming the reader with unnecessary information.

All three require citation. "In your own words" is not a substitute for credit — it just means you've done the cognitive work of restating, which earns you the right to weave the source's contribution into your own argument.

Common Knowledge and When Citation Is Unnecessary

Not every statement in an academic paper requires a citation. **Common knowledge** refers to information that is widely known, easily verified in multiple reliable sources, and unlikely to be disputed. Facts such as *the Earth orbits the Sun*, *Sacramento is the capital of California*, or *World War II ended in 1945* are considered common knowledge. Adding citations for these well-established facts generally does not strengthen your writing and can make it unnecessarily cluttered.

However, common knowledge depends on **your audience and your discipline**. A fact that is common knowledge among biology majors or nursing students may not be familiar to readers in a general education course. Likewise, specialized statistics, research findings, historical interpretations, or discipline-specific terminology usually require citations, even if they seem obvious to experts in the field.

When deciding whether to cite, ask yourself these questions:

- **Did I know this before I began my research?**
- **Would most readers in my intended audience recognize this as an undisputed fact?**
- **Can I find the same information presented consistently in multiple general reference sources, such as encyclopedias or textbooks?**

If you answer **no** to any of these questions, you should probably provide a citation. Likewise, if you are using a specific statistic, research finding, interpretation, or unique wording from another source, a citation is always required.

A good rule of thumb is simple: **when in doubt, cite**. Including an unnecessary citation is usually a minor style issue, but failing to acknowledge someone else's ideas or evidence can undermine your credibility and may constitute plagiarism.

APA 7 Quick Reference

APA 7 (American Psychological Association, 2020) is the citation style most common in the social sciences, education, business, nursing, and many community-college research-writing courses. The full APA Publication Manual covers thousands of cases; the basics below cover most undergraduate citations.

In-text citations

A **parenthetical citation** places the author's last name and publication year in parentheses at the end of a sentence.

- (Smith, 2021) at the end of the sentence.

A **narrative citation** incorporates the author's name into the sentence itself.

- Smith (2021) argued that...

Direct quotations also include a page number.

- (Smith, 2021, p. 47).

For works with **two authors**, include both names. For works with **three or more authors**, list the first author followed by *et al.*

Two authors: (Smith & Lopez, 2022)

- Three or more authors; (Smith et al., 2023)

Reference list

The reference list begins on a new page titled **References**. Entries are arranged alphabetically, double-spaced, and formatted with a hanging indent.

- **Journal article:** Author, A. A. (Year). Title of article. Journal Name, Volume(Issue), pages. <https://doi.org/xxx>
- **Book:** Author, A. A. (Year). Title of book. Publisher.
- **Book chapter:** Author, A. A. (Year). Title of chapter. In E. Editor (Ed.), Title of book (pp. xx–xx). Publisher.
- **Web page:** Author, A. A. (Year, Month Day). Title of page. Site Name. URL
- **Government report:** Agency Name. (Year). Title of report (Report No. XX). Publisher. URL

APA Guide

Always consult the official [APA guidelines](#) whenever you encounter an unfamiliar source type.

Citation Tools

Citation generators and reference managers such as Zotero, Mendeley, EndNote, and database citation tools can save considerable time, but they should never be trusted blindly.

Automatically generated citations frequently contain errors involving capitalization, italics, author names, publication dates, or DOIs. Treat these tools as helpful starting points rather than final products.

A useful rule is simple:

Use citation tools to create the first draft of a citation. Use your own judgment to proofread the final version.

Learning to recognize citation errors is part of becoming an independent researcher.

Conclusion

Mastering citation practices is the first step toward becoming an ethical researcher. As you begin creating presentations, websites, videos, podcasts, and other public-facing work, additional questions emerge about copyright, fair use, Creative Commons licensing, and the responsible use of artificial intelligence. The next chapter expands your understanding of information ethics beyond citation to explore the legal and ethical responsibilities that accompany creating and sharing information.

Chapter Summary

Academic integrity is the foundation of trustworthy scholarship. Ethical researchers acknowledge the work of others through accurate citation while contributing their own analysis and ideas. Understanding the different forms of plagiarism, knowing when to quote, paraphrase, or summarize, recognizing what qualifies as common knowledge, and applying APA citation principles all help ensure that your work is both honest and credible. Although citation tools can improve efficiency, they do not replace careful proofreading or critical judgment. Developing these habits now will prepare you for more advanced research and professional writing.

Key Concepts and Vocabulary

- **Academic integrity.** The commitment to honesty, transparency, and ethical behavior in academic work.
- **Citation.** A reference that identifies the source of borrowed ideas, words, or information.
- **Common knowledge.** Information that is widely known and generally does not require citation.
- **Direct plagiarism.** Copying another person's words without quotation marks or citation.
- **Mosaic plagiarism.** Combining language or ideas from sources without proper attribution.
- **Accidental plagiarism.** Failing to cite sources because of poor note-taking or inadequate paraphrasing.
- **Self-plagiarism.** Reusing previously submitted work without permission or disclosure.
- **Quotation.** Using a source's exact words with quotation marks and a citation.
- **Paraphrase.** Restating a specific idea from a source in your own words while citing the original author.
- **Summary.** Condensing the main ideas of a longer work into a shorter explanation in your own words with citation.
- **APA 7.** The citation style published by the American Psychological Association and widely used in the social sciences.

Critical Thinking Exercises

Try It: Reflection Activities

Activity 6.1A — Quote, Paraphrase, Summarize

Pick one paragraph from a source you're using. Write three versions of how you might use it: one direct quotation with citation, one paraphrase with citation, and one summary with citation. Note in 100 words which version you'd actually use in your paper, and why.

Think About It: Discussion Questions

1. "Most plagiarism is accidental." What does that suggest about how to design good note taking habits?
2. How can effective note-taking reduce the likelihood of plagiarism during the writing process?

References

- American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). <https://doi.org/10.1037/0000165-000>
- Association of College and Research Libraries. (2015, February 9). *Framework for information literacy for higher education*. American Library Association. <https://www.ala.org/acrl/standards/ilframework>

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6.2: Information Ethics and Intellectual Property

Unit 6 Ethical Use & Communication of Research | Chapter 6.2

Learning Objectives

- Explain the relationship between copyright, intellectual property, and information ethics.
- Apply the four factors of fair use to common educational situations.
- Distinguish between copyright, plagiarism, and copyright infringement.
- Interpret the basic terms of Creative Commons licenses.
- Evaluate the ethical and appropriate use of generative AI in academic and professional settings.
- Demonstrate transparency when using AI and other digital tools in research and writing.

Introduction

Finding and evaluating information is only part of becoming information literate. Responsible researchers must also understand the ethical and legal responsibilities that come with creating, sharing, and reusing information. Every article, photograph, song, dataset, video, and AI-generated image has an owner, a license, or both. Knowing what you may legally use and what you should ethically acknowledge is an essential part of participating in today's information ecosystem.

This chapter explores the broader ethical landscape surrounding information. You will learn how copyright protects creative work, when fair use permits limited reuse, how Creative Commons licenses encourage sharing, and how generative AI raises new questions about authorship, transparency, and intellectual responsibility.

Information Ethics Beyond Citation

Academic integrity extends beyond giving proper credit. Ethical information use includes **respecting creators' legal rights**, understanding the conditions under which information may be shared, and recognizing your own responsibilities as both a consumer and a creator of information.

Think about the different ways you interact with information each day. You may download images for a presentation, remix music into a video, create graphics with AI, share memes on social media, or publish your own writing online. Each of these activities involves ethical decisions about ownership, permission, attribution, and responsible use.

Information ethics asks questions such as:

- **Who owns this work?**
- **May I reuse it?**
- **Under what conditions?**
- **Should I ask permission?**
- **How should I acknowledge the creator?**

Answering these questions thoughtfully helps build trust and respect within academic and professional communities.

Respecting Intellectual Property

When people create something original, whether it is a book, photograph, song, software program, or research article, they generally have legal rights over how that work may be used. These protections are known collectively as **intellectual property**.

Researchers encounter several forms of intellectual property, but **copyright** is by far the most relevant because it governs most of the books, articles, images, videos, and other materials used in academic work.

Types of Intellectual Property

This table compares the three primary forms of intellectual property protection—copyright, trademark, and patent—and identifies the types of creative works, brands, and inventions each protects.

Type	Protects	Examples

Type	Protects	Examples
Copyright	Original creative expression	Books, journal articles, photographs, music, films, software
Trademark	Brands, names, logos, and slogans	Nike®, Coca-Cola®, Apple®
Patent	New inventions and processes	Medical devices, manufacturing technologies, pharmaceuticals

While all three forms protect intellectual property, researchers most often work with **copyright**, which determines how creative works may be copied, shared, and reused.

Understanding Copyright

Copyright is a legal protection that automatically applies whenever an original creative work is fixed in a tangible form. Registration is not required for copyright to exist.

Copyright Gives Creators the Right To:

- reproduce their work
- distribute copies
- create adaptations or derivative works
- perform or display the work publicly
- authorize others to use it

Owning a copy of a work does **not** transfer these rights. Purchasing a textbook, downloading an article, or borrowing a book from the library allows you to read or use that copy, but it does not give you permission to reproduce, post, or redistribute it.

Copyright exists to encourage creativity by allowing creators to benefit from their work while eventually contributing it to the public domain.

Understanding Fair Use

Because education, research, criticism, and public discussion benefit society, U.S. copyright law allows certain limited uses of copyrighted works without permission under the doctrine of **fair use**.

Fair use is **not** a checklist or an automatic exemption. Instead, courts evaluate each situation by weighing four factors together.

The Four Factors of Fair Use

This table summarizes the four factors courts consider when evaluating whether a particular use of copyrighted material qualifies as fair use under U.S. copyright law.

Factor	Question to Ask
Purpose and Character	Is my use educational, nonprofit, or transformative rather than commercial?
Nature of the Original Work	Is the original work primarily factual or highly creative?
Amount Used	Am I using only the portion necessary for my purpose?
Effect on the Market	Could my use replace the original work or reduce its value?

No single factor determines whether a use qualifies as fair use. Courts consider **all four factors together**, and each situation is evaluated individually.

Applying Fair Use

The difference between acceptable and unacceptable use often depends on context.

This table contrasts examples that are generally more likely to qualify as fair use with examples that are less likely, illustrating how context influences fair use decisions.

More Likely Fair Use	Less Likely Fair Use
Quoting several sentences in a research paper with citation	Uploading an entire copyrighted textbook to a public website
Including one graph in a classroom presentation	Posting an entire journal article online for others to download
Using short excerpts to critique or analyze a work	Sharing complete movies, songs, or books without permission

These examples illustrate general principles rather than absolute rules. When the answer is uncertain, consult your instructor, librarian, or your institution's copyright guidance.

Using Creative Commons Materials

Some creators want others to build upon their work while still receiving credit. Rather than requiring users to request permission individually, they may publish their work under a **Creative Commons (CC)** license.

Creative Commons licenses do **not** eliminate copyright. Instead, they clearly communicate what others are allowed to do with the work.

Common Creative Commons License Terms

This table explains the most common Creative Commons license conditions and the permissions or restrictions associated with each symbol.

Symbol	Meaning
BY (Attribution)	Credit the creator whenever you use the work.
SA (ShareAlike)	License any adaptations under the same terms.
NC (NonCommercial)	Do not use the work for commercial purposes.
ND (NoDerivatives)	Do not modify or adapt the work.

These conditions are often combined into a single license. For example, a **CC BY-NC-SA** license allows others to reuse and adapt the work for noncommercial purposes, provided they credit the creator and share any adaptations under the same license.

Creative Commons resources are excellent choices for student presentations, websites, videos, and multimedia projects because the creator has already granted permission under clearly stated conditions.

Copyright Is Not the Same as Plagiarism

Students often confuse plagiarism with copyright infringement, but they address different issues.

This table distinguishes plagiarism from copyright infringement by comparing their ethical and legal dimensions, helping clarify how the two concepts differ and where they may overlap.

Plagiarism	Copyright Infringement
An ethical issue	A legal issue
Failing to give proper credit	Violating a copyright holder's exclusive rights
Focuses on honesty and attribution	Focuses on permission and legal use

Although these concepts sometimes overlap, they are not identical. For example, you might properly cite a copyrighted photograph but still violate copyright if you reproduce it without permission. Conversely, you might legally quote a passage under fair use but still commit plagiarism if you fail to acknowledge the original author.

Responsible researchers think about **both** responsibilities. They give appropriate credit to creators while also respecting the legal rights that protect intellectual property.

Artificial Intelligence and Information Ethics

Generative AI has changed how information is created, shared, and evaluated. AI systems can draft text, generate images, summarize articles, write computer code, and answer questions within seconds. These capabilities create new opportunities, but they also create new ethical responsibilities.

Researchers should remember several principles.

- First, **AI is not an authority**. Large language models predict likely text rather than verify facts. They may fabricate citations, invent quotations, or confidently present incorrect information.
- Second, **AI does not replace critical thinking**. Brainstorming, outlining, editing, or generating practice questions may support learning, but evaluating evidence, synthesizing research, and developing arguments remain the responsibility of the researcher.
- Third, **follow institutional policy**. Different instructors and institutions have different expectations regarding AI use. Some prohibit it entirely, while others encourage responsible use with disclosure.
- Finally, **practice transparency**. If AI contributed meaningfully to your work and your instructor or publisher requires disclosure, explain how you used it and what steps you took to verify the information. Transparency demonstrates intellectual honesty and helps maintain trust between writers and readers.

Ethical Questions for the Future

Technology continues to evolve faster than laws and institutional policies. Questions surrounding AI-generated art, ownership of training data, deepfakes, synthetic media, algorithmic bias, and digital privacy will continue to shape research, education, and society.

Rather than memorizing rules that may change, develop habits that remain constant:

- Respect creators.
- Verify information.
- Credit contributions.
- Follow applicable laws and institutional policies.
- Be transparent about your own work.

These habits will continue to serve you regardless of how technology changes.

Conclusion

Information literacy is not only about finding credible information. It is also about participating responsibly in an information society. As you continue your academic journey, you will increasingly become a creator of knowledge through research papers, presentations, multimedia projects, and professional communication. Understanding your ethical and legal responsibilities prepares you to contribute to that conversation with integrity.

Chapter Summary

Information ethics extends beyond citation to encompass the legal and moral responsibilities of creating, sharing, and reusing information. Copyright protects original creative works, while fair use provides limited exceptions that support education, scholarship, and public discussion. Creative Commons licenses make sharing easier by clearly communicating how works may be reused. Because plagiarism and copyright infringement address different issues, ethical researchers consider both attribution and legal permission. Generative AI introduces new challenges, but the core principles of responsible scholarship remain unchanged: verify information, respect creators, think critically, and be transparent about your methods.

Key Concepts and Vocabulary

- **Intellectual property and copyright.** Legal protections for creators' rights to their work.
- **Fair use.** The exception that permits limited educational and transformative uses of copyrighted work.
- **Creative Commons license.** An open license through which creators permit sharing and reuse under stated conditions.
- **Plagiarism.** Presenting someone else's work or ideas as your own: direct, mosaic, accidental, or self-plagiarism.
- **Common knowledge.** Widely known, easily verified facts that can be stated without citation.
- **DOI.** Digital object identifier: the permanent link that takes a reader straight to a scholarly work.

Critical Thinking Exercises

Try It: Reflection Activities

Activity 6.2A – Can I Use It?

For each of the following, decide whether additional permission is likely needed, whether fair use may apply, or whether a Creative Commons license would allow reuse. Explain your reasoning.

1. A photograph from a commercial news website in a classroom presentation.
2. A Creative Commons image used in a course blog.
3. An AI-generated illustration included in a research poster.
4. A graph reproduced from a government report.

Think About It: Discussion Questions

1. Why is it important to distinguish between what is **legal** and what is **ethical** when using information?
2. As AI-generated content becomes more common, what responsibilities do researchers have to maintain transparency and trust?

References

American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.).

Association of College and Research Libraries. (2015, February 9). *Framework for information literacy for higher education*.

American Library Association. <https://www.ala.org/acrl/standards/ilframework>

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6.3: Communicating Research- Presentations, Posters and Research Abstracts

Unit 6: Ethical Use & Communication of Research | Chapter 6.3

Learning Objectives

- Translate written research into a multimedia format for a non-specialist audience.
- Structure a research talk that holds attention from the opening through the conclusion.
- Design a research poster using clear visual hierarchy, effective white space, and meaningful data visualizations.
- Write a concise research abstract that summarizes a project in 150 to 250 words.

Introduction

Completing a research project is only part of the research process. Research creates knowledge, but communication allows that knowledge to reach other people. A carefully researched paper has little impact if no one understands its findings or recognizes why they matter.

Why Research Communication Matters

Researchers communicate their work in many formats. A written paper allows readers to examine evidence in detail. A presentation highlights the most important ideas for a live audience. A poster communicates visually and invites conversation. An abstract introduces a project in just a few sentences and helps readers decide whether they want to learn more.

Each format serves a different purpose, but all share the same goal: helping an audience understand your work. Effective researchers recognize that communicating research requires its own set of skills. Rather than simply shrinking a paper into slides or copying paragraphs onto a poster, they rethink how to present the information in ways that are clear, engaging, and memorable.

Audience First

The first question to ask before creating any presentation, poster, or abstract is simple: **Who is this for?**

Your audience determines nearly every communication decision you make. The amount of background information you provide, the vocabulary you choose, the evidence you emphasize, and even the examples you use should reflect what your audience already knows and what they need to learn.

Imagine presenting the same research project to three different groups. Your classmates may already understand the course concepts but need help following your specific argument. Faculty members may expect more methodological detail and stronger evidence. Members of the general public may know little about the topic and benefit from everyday language rather than specialized terminology.

Matching Your Message to Your Audience

Adapting Research to Different Audiences. Tailor your language, evidence, and emphasis to meet the needs of your audience.

Audience	Prior Knowledge	Best Approach	Emphasis
Classmates	Moderate	Explain ideas clearly with familiar course concepts	Main findings and significance
Faculty	High	Use discipline-specific language appropriately	Methods, evidence, and limitations
General public	Limited	Use plain language and define technical terms	Why the research matters
Decision makers	Mixed	Be concise and action oriented	Evidence supporting recommendations

Audience awareness also helps you decide what **not** to include. Your research paper may contain dozens of citations and pages of detailed analysis, but a presentation should focus only on the ideas your audience needs to understand your conclusions.

When in doubt, simplify your language without oversimplifying your ideas. Clear communication demonstrates mastery of a topic.

Structuring a Research Talk

Whether you have five minutes or thirty, effective presentations follow a clear organizational pattern that helps the audience understand and remember your message. For the literature review presentations in this course, your presentation should mirror the structure of your paper by introducing your research question, discussing the major themes that emerged from the literature, and concluding with the significance of your findings.

1. Hook

Begin by capturing your audience's attention with an engaging opening. A surprising statistic, a brief story, a thought-provoking question, a current event, or an unexpected comparison can all provide an effective hook. Your goal is to show why your topic matters and why your audience should care.

2. Research Question

State your research question clearly and early in the presentation. This question provides the framework for everything that follows and helps your audience understand the purpose of your research.

3. Major Themes

Organize the body of your presentation around **at least three major themes** that emerged from your literature review. Rather than discussing one source at a time, synthesize what multiple researchers found about each theme. Explain the main ideas, identify areas of agreement or disagreement, and show how the evidence connects to your research question.

4. Supporting Evidence

Support each theme with evidence from your sources. Include brief quotations, statistics, or visualizations only when they strengthen your explanation. The emphasis should remain on your analysis rather than on lengthy summaries of individual studies.

5. Conclusion

Conclude by bringing the themes together and answering the larger question: **What did the literature reveal?** Summarize the overall patterns you discovered, explain why they matter, and identify any important gaps or directions for future research. End with a clear takeaway that leaves your audience with the significance of your research rather than simply announcing that you are finished.

Designing an Effective Multi-Media Presentation

Well-designed slides support your presentation by reinforcing your message rather than competing with it. The goal is not to place your entire presentation on the screen, but to provide visual cues that help your audience follow your ideas. Clean, consistent slides keep the focus on your research instead of distracting viewers with unnecessary text or design elements.

As you create your slides, keep these principles in mind:

- **One main idea per slide.** Each slide should communicate a single concept or finding. If a slide feels crowded or tries to cover multiple topics, divide it into two slides.
- **Use large, readable text.** Your audience should be able to read your slides from the back of the room. Titles should stand out clearly, and body text should generally be at least 28-point font.
- **Keep text to a minimum.** Use keywords and short phrases instead of complete paragraphs. Your spoken explanation should provide the details while the slides highlight the main points.
- **Choose meaningful visuals.** Charts, graphs, diagrams, photographs, and illustrations should help explain your findings, not simply decorate the presentation.
- **Maintain a consistent design.** Using the same fonts, colors, and layout throughout the presentation creates a professional appearance and allows your audience to focus on your content rather than changes in formatting.

Limit Text

If your audience can read everything on the slide, they no longer need you. Slides should reinforce your message, not replace it. Instead of paragraphs, use bullet phrases. Stick to one consistent color palette and large, readable text.

Deliver the Presentation, Don't Read It

One of the most common presentation mistakes is reading directly from the slides. Slides are meant to support your presentation, not replace it. If every word appears on the screen, your audience will read instead of listening to you. Instead, think of each slide as a prompt that highlights your main ideas with short phrases, images, graphs, or diagrams while you provide the explanation.

Speaking conversationally also helps you appear more confident and engaged. Rather than memorizing every sentence, become familiar enough with your research to explain it naturally in your own words.

Designing a Research Poster

Unlike an oral presentation, a research poster must communicate its message even when the author is not actively explaining it. Conference attendees often spend less than two minutes scanning a poster before deciding whether to stop and ask questions, so the main idea should be immediately apparent. Rather than thinking of a poster as a miniature research paper, think of it as a visual conversation starter that highlights your most important findings.

Effective research posters share three key design principles:

- **Visual hierarchy:** Arrange information so readers naturally know where to look first. The title should be the largest text on the poster, followed by section headings and then body text. A clear layout guides readers smoothly from one section to the next.
- **White space:** Empty space is an important design element, not wasted space. It separates ideas, improves readability, and prevents the poster from feeling crowded or overwhelming.
- **Data visualization:** Use graphs, charts, diagrams, photographs, or other visuals to communicate findings quickly. A well-designed figure should allow readers to grasp its main point within a few seconds.

A typical research poster includes a title and author information, an introduction or background section, the research question, methods, findings, conclusions, references, and contact information or a QR code. Of these sections, the findings are usually the largest and most visually prominent because they represent the primary contribution of the research.

Choosing the Right Visual

Different types of visuals communicate information in different ways. Choosing the right visual can make your findings easier to understand and help your audience quickly recognize the most important points. Consider using the following types of visuals based on the information you want to present:

- **Bar charts** compare categories or groups.
- **Line graphs** show changes or trends over time.
- **Tables** present exact values or detailed numerical information.
- **Photographs** illustrate an object, location, person, or process.
- **Diagrams** explain relationships, systems, or workflows.
- **Infographics** combine text, visuals, and data to summarize multiple ideas in a clear, engaging format.

Select the simplest visual that communicates your message accurately and supports your conclusions.

Typical Poster Sections

Although formats vary by discipline, most research posters include:

- Title and author information
- Brief introduction or background
- Research question
- Methods
- Findings
- Conclusions

- References
- Contact information or QR code

The findings section is typically the largest and most visually prominent area of the poster.

Poster Design Checklist

Before printing your poster, ask yourself:

- Is the title readable from several feet away?
- Can someone identify the main finding within a few seconds?
- Are figures clearly labeled?
- Is there enough white space?
- Are fonts consistent?
- Have you proofread every section?

Writing a Research Abstract

A research abstract is a concise, standalone summary of your project that allows readers to quickly understand what you studied, how you conducted the research, what you found, and why it matters. Most abstracts range from **150 to 250 words**, depending on the requirements of a journal, conference, or course assignment. Because reviewers often read dozens or even hundreds of abstracts, yours should communicate the essential information clearly and efficiently.

Although disciplines vary slightly, most effective abstracts include the same five elements:

- **Context:** Introduce the broader topic and explain why it is important.
- **Research Question:** Clearly state the question or problem your project investigated.
- **Methods:** Briefly describe how you conducted the research.
- **Findings:** Summarize your most important results or conclusions.
- **Implications:** Explain why the findings matter and what they contribute to the conversation or future research.

Following this structure helps readers quickly understand the purpose, approach, and significance of your research.

Editing an Abstract

Writing a concise abstract requires careful editing. Every word should serve a purpose, so remove unnecessary language and replace vague statements with specific descriptions of your research. Because an abstract is a brief summary rather than a full paper, avoid quotations, citations, and lengthy background information. Focus instead on clearly communicating your research question, methods, findings, and conclusions in as few words as possible. Every sentence should move the reader toward understanding your project.

Instead of writing:

It is important to note that social media has become increasingly popular among young adults in recent years...

Write:

This study examined how social media use influences sleep quality among community college students.

Strong vs. Weak Abstracts

Weak vs. Strong Research Abstracts. Key differences between ineffective and effective abstracts.

Weak Abstract	Strong Abstract
Begins with lengthy background	Introduces the problem quickly
Uses vague language	States the research question clearly
Describes every method	Includes only essential methods
Reports general conclusions	Presents specific findings

Weak Abstract

Ends abruptly

Strong Abstract

Explains the significance of the work

 Research Tip

Read your abstract aloud before submitting it. If a sentence feels awkward to say, it will likely be awkward to read.

Presenting at Undergraduate Conferences

Many colleges and universities sponsor undergraduate research conferences, and regional or national organizations frequently welcome student scholarship.

Submitting your work usually requires:

- a research abstract
- a short author biography
- faculty sponsorship or endorsement (for some conferences)

Presenting your research allows you to receive feedback from scholars outside your classroom, build confidence as a communicator, and strengthen applications for graduate school, scholarships, or employment.

Even if your project began as a course assignment, it may be valuable enough to share beyond your campus.

Public Speaking, Nerves, and Practice

Feeling nervous before a presentation is normal. Even experienced researchers often feel anxious before speaking. The goal is not to eliminate those nerves but to prepare well enough that they no longer control your performance.

The best preparation is practice. Rehearse your presentation out loud rather than silently reading your notes, and use a timer to make sure you stay within your allotted time. During the presentation, speak slowly, pause briefly between major ideas, and make eye contact with different parts of the room. These simple habits help your audience follow your message and make you appear more confident.

If you lose your place or stumble over a word, pause, collect your thoughts, and continue. Most audiences notice mistakes far less than presenters imagine. Remember that your audience wants you to succeed. They are there to learn about your research, not to judge a perfect performance.

 Presentation Tip

Practice until you know your ideas, not your exact wording. Understanding your research allows you to adapt naturally if you forget a sentence or receive an unexpected question.

Conclusion

Every research project begins with curiosity, but it reaches its full value only when it is shared. Whether you submit a paper, present at a conference, design a poster, or explain your work to classmates, you become part of the scholarly conversation. Effective communication allows your ideas to inform others, invites meaningful feedback, and encourages new questions that continue the cycle of research.

Chapter Summary

Research communication is a skill that extends beyond writing a paper. Every presentation, poster, and abstract should begin with a clear understanding of the intended audience and what that audience needs to know. Effective presentations follow a logical structure that moves from an engaging introduction to a clear explanation of the research question, methods, findings, and significance. Well-designed slides and posters emphasize visual hierarchy, readability, and meaningful graphics rather than large amounts of text. Research abstracts summarize an entire project in just a few carefully chosen sentences, making concise writing

essential. Whether you share your work in class or at an undergraduate conference, thoughtful preparation and practice allow your research to inform, persuade, and inspire others.

Key Concepts and Vocabulary

- **Audience analysis.** Determining who will receive your work and what they need before you build anything.
- **Five-part talk.** Hook, question, methods, findings, so what: the structure of an effective research talk.
- **Visual hierarchy.** Sizing and placement that let a reader follow a poster at a glance.
- **White space.** The empty areas that make the rest of a poster legible.
- **Research abstract.** A 150 to 250 word standalone summary covering context, question, methods, findings, and implications.

Critical Thinking Activities

? Try It: Reflection Activities

Activity 6.2A — Draft a Research Abstract

Write a 250-word abstract for your research paper, covering context, question, methods, findings, and implications. Edit it twice: once to tighten language, once to cut anything not essential.

Activity 6.2B — Build a Five-Slide Deck

Create a five-slide deck for a 5-minute talk on your research, following the hook → question → methods → findings → so-what structure.

Activity 6.2C — Sketch a Poster

Sketch a one-page poster mockup for your research project. Don't worry about polish; focus on hierarchy, white space, and where each section will go. Label your sections, indicate where visuals would go, and submit the sketch.

? Think About It: Discussion Questions

1. Why is "audience first" the right starting point for any research presentation? What goes wrong when you skip it?
2. What's the hardest part of compressing 20 pages of research into a 250-word abstract — for you?

References

Association of College and Research Libraries. (2015, February 9). *Framework for information literacy for higher education*. American Library Association. <https://www.ala.org/acrl/standards/ilframework>

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